WITTEN–PARKER–TAUBES PROOF OF THE POSITIVE ENERGY THEOREM

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Abstract. We give a self-contained exposition of the spinorial proof of the positive energy theorem, following the argument introduced by Witten and its analytic completion by Parker and Taubes. Starting from an asymptotically flat initial data set (M^3, g, h) satisfying the dominant energy condition, we define the hypersurface Dirac operator and derive Witten's Weitzenböck identity. Its integrated form yields a nonnegative bulk term and a boundary flux at spatial infinity. The analytic input is the existence and uniqueness of solutions to the Witten equation $\mathscr{D}\psi = 0$ with prescribed asymptotics, obtained using weighted Sobolev spaces and Fredholm theory on asymptotically flat ends. Combining these ingredients gives $E \geq |P|$ on each end.

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1. The positive energy theorem and asymptotically flat initial data

Let $(N, g_{\alpha\beta})$ be a 4-dimensional Lorentz manifold of signature $(-,+,+,+)$, and let

$$M \subset N$$

be a complete oriented spacelike hypersurface. We denote by the same symbol g the induced Riemannian metric on M. Greek indices α, β, μ, ν run from 0 to 3, while Latin indices i, j, k, ℓ run from 1 to 3. Along M we choose a future-pointing unit normal vector field e_0, and when convenient we extend it to an adapted orthonormal frame

$$\{e_0, e_1, e_2, e_3\}$$

with e_1, e_2, e_3 tangent to M.

The aim of this section is to fix the geometric hypotheses and the asymptotic quantities that enter the positive energy theorem. The point is not to discuss the full Cauchy problem for Einstein's equation, but rather to isolate exactly the data that appear in Witten's argument and in the rigorous formulation of Parker and Taubes.

1.1. Spacelike hypersurfaces and initial data.

Since M is spacelike, the restriction of the Lorentz metric to TM is positive definite, so (M, g) is a Riemannian 3-manifold. The extrinsic geometry of the inclusion $M \hookrightarrow N$ is encoded by its second fundamental form.

Definition 1.1. The *second fundamental form* of the hypersurface $M \subset N$ with respect to the chosen future-pointing unit normal e_0 is the symmetric tensor h_{ij} on M defined by

$$h_{ij} = g(\nabla^N_{e_i} e_0, e_j), \tag{1.2}$$

where ∇^N is the Levi-Civita connection of the spacetime metric.

It is useful to record the basic identities associated to (1.2). Since $g(e_0, e_0) = -1$, differentiating this relation in the direction of any tangent vector e_i gives

$$0 = e_i\, g(e_0, e_0) = 2\, g(\nabla^N_{e_i} e_0, e_0),$$

hence $\nabla^N_{e_i} e_0$ is tangent to M. Therefore the right-hand side of (1.2) really defines a tensor on M.

Moreover, if X, Y are tangent vector fields on M, then metric compatibility and the torsion-free property of ∇^N imply

$$g(\nabla^N_X e_0, Y) = g(\nabla^N_Y e_0, X),$$

so $h_{ij} = h_{ji}$. It follows that the ambient Levi-Civita connection splits along M as

$$\nabla^N_{e_i} e_j = \nabla_{e_i} e_j - h_{ij} e_0, \qquad \nabla^N_{e_i} e_0 = h_{ij} e_j, \tag{1.3}$$

where ∇ denotes the Levi-Civita connection of the induced metric g on M. The first formula is obtained by decomposing $\nabla^N_{e_i} e_j$ into tangent and normal parts and then computing its normal component:

$$g(\nabla^N_{e_i} e_j, e_0) = -g(e_j, \nabla^N_{e_i} e_0) = -h_{ij}.$$

The second formula is just the definition of h_{ij} written in the basis $\{e_j\}$ of TM.

Thus an initial data set for the theorem consists of the pair

$$(M, g, h),$$

where g is the induced Riemannian metric on the spacelike hypersurface and h is its second fundamental form.

1.2. **Asymptotically flat ends.** The positive energy theorem concerns isolated gravitating systems. Mathematically, the isolated character is expressed by the requirement that outside a compact set the initial hypersurface break up into finitely many ends, each of which approaches Euclidean space with a definite rate.

Definition 1.4. We say that the complete initial data set (M, g, h) is *asymptotically flat* if there exists a compact set $K \subset M$ such that

$$M \setminus K = M_1 \sqcup \cdots \sqcup M_k$$

is a finite disjoint union of open sets, called the *ends* of M, with the following property.

For each end M_ℓ there is a diffeomorphism from M_ℓ onto the complement of a contractible compact set in $\mathbb{R}^3$. Using the standard coordinates

$$x = (x^1, x^2, x^3)$$

on $\mathbb{R}^3$ and writing

$$r = |x| = \big((x^1)^2 + (x^2)^2 + (x^3)^2\big)^{1/2},$$

the induced metric on M_ℓ has the form

$$g_{ij} = \delta_{ij} + a_{ij}, \tag{1.5}$$

where

$$a_{ij} = O(r^{-1}), \qquad \partial_k a_{ij} = O(r^{-2}), \qquad \partial_\ell \partial_k a_{ij} = O(r^{-3}), \tag{1.6}$$

and the second fundamental form satisfies

$$h_{ij} = O(r^{-2}), \qquad \partial_k h_{ij} = O(r^{-3}). \tag{1.7}$$

Here and below, every derivative in (1.6) and (1.7) is taken with respect to the Euclidean coordinates on the chosen end. We shall often identify the end M_ℓ with its image in $\mathbb{R}^3$ and use the Euclidean background quantities attached to these coordinates without further comment.

The decay rates in Definition 1.4 are exactly the ones needed in the original spinorial proof. The first derivatives of the metric appear in the flux integral defining the total energy, so they must be $O(r^{-2})$. Likewise, h_{ij} itself appears in the flux

integral defining the total momentum, so it must also be $O(r^{-2})$. The extra derivative bounds are used to control error terms and to justify the limiting procedures at infinity.

It is convenient to write the Euclidean area element on the coordinate sphere

$$S_r = \{x \in \mathbb{R}^3 : |x| = r\}$$

in two interchangeable ways. If ν denotes the outward Euclidean unit normal to S_r and dS its Euclidean surface measure, then

$$d\Omega^i := \nu^i \, dS$$

is the standard Euclidean vector-valued surface element. In coordinates this is also

$$d\Omega^i = \partial_i \lrcorner (dx^1 \wedge dx^2 \wedge dx^3).$$

We will use both notations according to convenience.

The asymptotic assumptions immediately imply the size of the energy and momentum integrands.

Lemma 1.8. *On an asymptotically flat end, one has*

$$\partial_j g_{ij} - \partial_i g_{jj} = O(r^{-2}), \tag{1.9}$$

and

$$h_{ik} - \delta_{ik} h_{jj} = O(r^{-2}). \tag{1.10}$$

Consequently, the corresponding flux integrals over S_r are uniformly bounded as $r \to \infty$.

Proof. Since $g_{ij} = \delta_{ij} + a_{ij}$ and δ_{ij} is constant, we have

$$\partial_j g_{ij} - \partial_i g_{jj} = \partial_j a_{ij} - \partial_i a_{jj}.$$

By (1.6), each first derivative of a_{ij} is $O(r^{-2})$, and (1.9) follows.

Similarly, (1.7) gives $h_{ik} = O(r^{-2})$, while

$$h_{jj} = \delta^{jj'} h_{jj'}$$

is a finite sum of terms of order $O(r^{-2})$. Hence

$$h_{ik} - \delta_{ik} h_{jj} = O(r^{-2}),$$

which is (1.10).

Finally, the Euclidean area of S_r is $4\pi r^2$, so integrating an $O(r^{-2})$ quantity over S_r gives a quantity of order $O(1)$. □

The lemma only gives boundedness. Existence of the limits defining energy and momentum uses the additional derivative information in Definition 1.4 together with the field equations.

1.3. **ADM energy and momentum.** We now define the total energy and total momentum of an asymptotically flat end. These are flux integrals at spatial infinity. Their form is dictated by the Hamiltonian formulation of general relativity, but for us the relevant point is simply that they are quantities depending only on the initial data (g, h) on M.

Definition 1.11. Let M_ℓ be an asymptotically flat end.

(i) The *ADM energy* of the end M_ℓ is

$$E_\ell = \frac{1}{16\pi} \lim_{r\to\infty} \int_{S_r} \big(\partial_j g_{ij} - \partial_i g_{jj}\big)\, d\Omega^i. \tag{1.12}$$

(ii) The *ADM momentum* of the end M_ℓ is the vector

$$P_\ell = \big((P_\ell)_1, (P_\ell)_2, (P_\ell)_3\big) \in \mathbb{R}^3$$

whose k-th component is

$$(P_\ell)_k = \frac{1}{8\pi} \lim_{r\to\infty} \int_{S_r} \big(h_{ik} - \delta_{ik} h_{jj}\big)\, d\Omega^i. \tag{1.13}$$

Since $d\Omega^i = \nu^i dS$, the same formulas can be written as

$$E_\ell = \frac{1}{16\pi} \lim_{r\to\infty} \int_{S_r} \big(\partial_j g_{ij} - \partial_i g_{jj}\big)\nu^i\, dS,$$

and

$$(P_\ell)_k = \frac{1}{8\pi} \lim_{r\to\infty} \int_{S_r} \big(h_{ik} - \delta_{ik} h_{jj}\big)\nu^i\, dS.$$

The vector-valued notation in (1.12)–(1.13) is often slightly cleaner when computing with differential forms, while the normal-vector notation is closer to the usual flux interpretation.

Let us also note why the momentum formula is written using the Euclidean tensor δ_{ij} on the chosen end. Since

$$g_{ij} = \delta_{ij} + O(r^{-1}), \qquad g^{ij} = \delta^{ij} + O(r^{-1}),$$

and since $h_{ij} = O(r^{-2})$, we have

$$\mathrm{tr}_g h - \mathrm{tr}_\delta h = (g^{ij} - \delta^{ij})h_{ij} = O(r^{-1})O(r^{-2}) = O(r^{-3}).$$

Therefore

$$\big(\mathrm{tr}_g h\big) g_{ik} - \big(\mathrm{tr}_\delta h\big)\delta_{ik} = O(r^{-3}),$$

and replacing the Euclidean expression in (1.13) by the corresponding expression with g changes the integrand by $O(r^{-3})$. Its flux through S_r is then $O(r^{-1})$, which vanishes as $r \to \infty$. Thus the asymptotic value of the momentum depends only on the end, not on this choice of writing.

Remark 1.14. The energy and momentum are defined separately on each end. This is important because the positive energy theorem is an inequality *on each end*, not merely a statement about the sum over all ends.

1.4. **The dominant energy condition.** The theorem is proved under a positivity hypothesis on matter. We state it in the form most closely adapted to the spinorial argument.

Let $T_{\alpha\beta}$ be a symmetric tensor field on the spacetime N satisfying Einstein's equation

$$R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R = 8\pi T_{\alpha\beta}. \tag{1.15}$$

At a point $p \in M$, choose an adapted orthonormal frame

$$\{e_0, e_1, e_2, e_3\}$$

with e_0 normal to M and e_1, e_2, e_3 tangent to M, and set

$$T_{00} = T(e_0, e_0), \qquad T_{0i} = T(e_0, e_i).$$

Definition 1.16. We say that the *dominant energy condition* holds along M if for every point $p \in M$ and every future-pointing timelike unit vector e_0 normal to M, the covector $T(e_0, \cdot)$ is non-spacelike and future-pointing. Equivalently, in any adapted orthonormal frame one has

$$T_{00} \geq \left(-T_{0i}T_0{}^i\right)^{1/2}. \tag{1.17}$$

Since the metric has signature $(-,+,+,+)$, the spatial part of the norm is positive definite, so

$$-T_{0i}T_0{}^i = \sum_{i=1}^{3}(T_{0i})^2.$$

Thus (1.17) can be written more concretely as

$$T_{00} \geq \left(\sum_{i=1}^{3}(T_{0i})^2\right)^{1/2}. \tag{1.18}$$

In particular,

$$T_{00} \geq |T_{0i}| \qquad \text{for each } i = 1, 2, 3.$$

Later, when the square of the hypersurface Dirac operator is computed, the zero-order term will involve exactly the combination

$$T_{00} + T_{0i}\, e^0 \cdot e^i,$$

and the inequality (1.18) is what makes that term nonnegative.

1.5. **Statement of the theorem.** We can now state the theorem in the form that will be proved in this exposition.

Theorem 1.19 (Positive energy theorem)**.** *Let $(N, g_{\alpha\beta})$ be a 4-dimensional Lorentz manifold satisfying Einstein's equation (1.15), and let*

$$M \subset N$$

be a complete oriented spacelike hypersurface. Assume that:

(i) the induced initial data set (M, g, h) is asymptotically flat in the sense of Definition 1.4;
(ii) the dominant energy condition of Definition 1.16 holds along M.

Then for each end M_ℓ of M,

$$E_\ell \geq |P_\ell|, \tag{1.20}$$

where E_ℓ and P_ℓ are the ADM energy and momentum defined by (1.12) and (1.13).

The rest of the paper is devoted to a spinorial proof of Theorem 1.19. The central idea is to solve a Dirac equation on the hypersurface M with prescribed constant behavior on each end, square the corresponding Dirac operator, and identify the resulting boundary term with the quantity

$$E_\ell \langle \psi_\infty, \psi_\infty \rangle + \left\langle \psi_\infty,\, P_{\ell k}\, e^0 \cdot dx^k \cdot \psi_\infty \right\rangle.$$

The inequality (1.20) then follows by choosing the constant spinor ψ_∞ appropriately.

2. Spinors along a spacelike hypersurface

The spinorial proof of Theorem 1.19 uses spinors defined *along* the spacelike hypersurface $M \subset N$. The relevant bundle is not the usual 3-dimensional spinor bundle built solely from the Riemannian geometry of (M, g), although that geometry does enter. Instead, one starts with the $3+1$-dimensional Dirac representation and restricts it to M, where the future-pointing unit normal e_0 supplies the extra structure needed to obtain a positive definite Hermitian metric.

We begin with the linear algebra of Dirac spinors in Minkowski space and then pass to the corresponding vector bundle over M.

2.1. Dirac spinors in dimension $3+1$.

Let

$$V = \mathbb{C}^2$$

be the fundamental representation of $SL(2,\mathbb{C})$. This complex vector space carries an $SL(2,\mathbb{C})$-invariant symplectic form, which we denote by a. Since a is nondegenerate, every linear map

$$A : V^* \longrightarrow V$$

has an a-adjoint

$$A' : V \longrightarrow V^*$$

characterized by

$$a(\lambda, A'\xi) = a(A\lambda, \xi) \qquad \text{for all } \lambda \in V^*,\ \xi \in V.$$

Minkowski space $\mathbb{R}^{3,1}$ may be realized as a real 4-dimensional subspace of $\mathrm{Hom}(V^*, V)$. Choosing a basis of V, this subspace is identified with the Hermitian 2×2 matrices via

$$x = (x^0, x^1, x^2, x^3) \longmapsto X(x) = \begin{pmatrix} x^0 + x^1 & x^2 + ix^3 \\ x^2 - ix^3 & x^0 - x^1 \end{pmatrix}. \tag{2.1}$$

A direct computation gives

$$\det X(x) = -(x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2, \tag{2.2}$$

so the determinant recovers the Lorentz quadratic form of signature $(- + ++)$.

Definition 2.3. The *Dirac spinor space* in dimension $3+1$ is

$$S_{3,1} := V \oplus V^*.$$

For $x \in \mathbb{R}^{3,1} \subset \mathrm{Hom}(V^*, V)$, Clifford multiplication by x is the linear map

$$x\cdot : S_{3,1} \longrightarrow S_{3,1}$$

defined by

$$x \cdot (\xi, \eta) := \big(x\eta,\ x'\xi\big), \qquad (\xi, \eta) \in V \oplus V^*. \tag{2.4}$$

The point of (2.4) is that the off-diagonal form of $x\cdot$ encodes the Lorentz metric.

Lemma 2.5. *For $x, y \in \mathbb{R}^{3,1}$, Clifford multiplication satisfies*

$$x \cdot y \cdot + y \cdot x\cdot = -2\, g(x, y)\, \mathrm{Id}_{S_{3,1}}. \tag{2.6}$$

In particular,

$$x \cdot x\cdot = -g(x, x)\, \mathrm{Id}_{S_{3,1}}. \tag{2.7}$$

Proof. We first compute $x \cdot x\cdot$. From (2.4),

$$x \cdot x \cdot (\xi, \eta) = x \cdot (x\eta, x'\xi) = (xx'\xi,\ x'x\eta).$$

It is therefore enough to identify the endomorphisms xx' of V and $x'x$ of V^*.

We claim that $x'x$ is a scalar multiple of the identity on V^*. Indeed, for any $\lambda \in V^*$,

$$a(\lambda, (x'x)\lambda) = a(x\lambda, x\lambda) = 0,$$

because a is skew-symmetric. On a 2-dimensional symplectic vector space, an endomorphism B with $a(\lambda, B\lambda) = 0$ for every λ must be a scalar multiple of the identity. Hence

$$x'x = \beta \operatorname{Id}_{V^*}$$

for some $\beta \in \mathbb{C}$. The same argument gives

$$xx' = \beta \operatorname{Id}_V.$$

Taking determinants, we obtain

$$\beta^2 = \det(x'x) = (\det x)^2.$$

Thus $\beta = \pm \det x$. To determine the sign, evaluate at the matrix

$$x = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

which corresponds, under (2.1), to the vector $(1, 0, 0, 0)$. For this x, one checks directly from (2.4) that

$$x \cdot x\cdot = \operatorname{Id}_{S_{3,1}},$$

while $\det x = 1$ and $g(x, x) = -1$. Hence

$$x'x = xx' = -\det x \operatorname{Id},$$

and therefore in general

$$x'x = xx' = -g(x, x) \operatorname{Id}.$$

Substituting into the formula for $x \cdot x\cdot$ gives (2.7).

Finally, (2.6) follows from polarization:

$$(x + y) \cdot (x + y)\cdot = x \cdot x \cdot + x \cdot y \cdot + y \cdot x \cdot + y \cdot y\cdot = -g(x + y, x + y)\mathrm{Id}.$$

Subtracting the identities for $x \cdot x\cdot$ and $y \cdot y\cdot$ yields (2.6). □

2.2. **Two Hermitian forms.** The Dirac representation carries a natural Hermitian form which is invariant under $SL(2, \mathbb{C})$. This form is not positive definite, and the positive definite structure needed later is obtained only after the choice of a timelike covector.

Proposition 2.8. *There is an $SL(2, \mathbb{C})$-invariant Hermitian form*

$$(\,\cdot\,,\,\cdot\,)$$

on $S_{3,1}$ with the property that for every $x \in \mathbb{R}^{3,1}$,

$$(x \cdot \psi, \phi) = (\psi, x \cdot \phi), \qquad \psi, \phi \in S_{3,1}. \tag{2.9}$$

This form is nondegenerate but not positive definite.

Proof. This is the standard invariant Hermitian form on the Dirac representation $S_{3,1} = V \oplus V^*$. The key property for us is (2.9), namely that Clifford multiplication by a real vector is Hermitian with respect to $(\,\cdot\,,\,\cdot\,)$. The form is not positive definite; otherwise every operator $x\cdot$ with $g(x,x) > 0$ would have nonnegative square, contradicting (2.7). □

Now fix a future-pointing unit timelike covector e^0. Since $g(e^0, e^0) = -1$, Lemma 2.5 gives

$$(e^0\cdot)^2 = \mathrm{Id}_{S_{3,1}}.$$

Moreover, $e^0\cdot$ exchanges the two summands V and V^*, so it identifies V^* with V. This is the extra structure supplied by the choice of time direction.

Definition 2.10. The *positive definite Hermitian inner product* on $S_{3,1}$ associated to e^0 is

$$\langle \psi, \phi \rangle := (e^0 \cdot \psi, \phi), \qquad \psi, \phi \in S_{3,1}. \tag{2.11}$$

Lemma 2.12. *With respect to $\langle\,\cdot\,,\,\cdot\,\rangle$, the following hold.*

(i) $\langle\,\cdot\,,\,\cdot\,\rangle$ is positive definite.
(ii) $e^0\cdot$ is Hermitian:

$$\langle e^0 \cdot \psi, \phi \rangle = \langle \psi, e^0 \cdot \phi \rangle.$$

(iii) If $x \in \mathbb{R}^{3,1}$ satisfies $g(x, e^0) = 0$, then $x\cdot$ is skew-Hermitian:

$$\langle x \cdot \psi, \phi \rangle = -\langle \psi, x \cdot \phi \rangle.$$

Proof. We first prove (ii). Using (2.11), (2.9), and $(e^0\cdot)^2 = \mathrm{Id}$, we have

$$\langle e^0 \cdot \psi, \phi \rangle = (e^0 \cdot e^0 \cdot \psi, \phi) = (\psi, \phi),$$

while

$$\langle \psi, e^0 \cdot \phi \rangle = (e^0 \cdot \psi, e^0 \cdot \phi) = (\psi, e^0 \cdot e^0 \cdot \phi) = (\psi, \phi).$$

So $e^0\cdot$ is Hermitian.

Next, if $g(x, e^0) = 0$, then Lemma 2.5 gives

$$e^0 \cdot x \cdot + x \cdot e^0\cdot = 0.$$

Hence

$$\langle x \cdot \psi, \phi \rangle = (e^0 \cdot x \cdot \psi, \phi) = (-x \cdot e^0 \cdot \psi, \phi) = -(e^0 \cdot \psi, x \cdot \phi) = -\langle \psi, x \cdot \phi \rangle,$$

which proves (iii).

Finally, we prove (i). Since $e^0\cdot : V^* \to V$ is an isomorphism, after using it to identify V^* with V the spinor space becomes

$$S_{3,1} \cong V \oplus V.$$

In a basis adapted to this identification, $e^0\cdot$ acts as the identity on each summand, and $\langle\,\cdot\,,\,\cdot\,\rangle$ becomes the standard positive Hermitian inner product on $V \oplus V$. Thus

$$\langle \psi, \psi \rangle > 0 \qquad \text{for every } \psi \neq 0,$$

as required. □

Later all coercive estimates and all positivity statements will be made with respect to $\langle\,\cdot\,,\,\cdot\,\rangle$, not with respect to the indefinite form $(\,\cdot\,,\,\cdot\,)$.

2.3. **Spin structures on M and constant spinors at infinity.** We now globalize the preceding linear algebra.

Let $F_{\mathrm{SO}}(M) \to M$ be the principal $SO(3)$-bundle of oriented orthonormal frames of TM. A spin structure on M is a principal Spin(3)-bundle

$$P_{\mathrm{Spin}}(M) \longrightarrow F_{\mathrm{SO}}(M)$$

lifting this frame bundle.

Proposition 2.13. *Every oriented 3-manifold admits a spin structure.*

Proof. This is a standard theorem: every orientable 3-manifold is spin, equivalently $w_2(M) = 0$. We choose one such spin structure once and for all. □

Since

$$\mathrm{Spin}(3) \cong SU(2) \subset SL(2,\mathbb{C}) \cong \mathrm{Spin}^+(3,1),$$

the chosen Spin(3)-bundle on M also defines a Spin(3, 1)-bundle along M. We therefore obtain the associated complex vector bundle

$$S := P_{\mathrm{Spin}}(M) \times_{\widetilde{\rho}} S_{3,1}, \tag{2.14}$$

where $\widetilde{\rho}$ is the restriction of the Dirac representation of Spin(3, 1) to Spin(3). This is the *Dirac spinor bundle along M*. Its sections are the spinor fields that appear in the positive energy theorem.

Because S is associated to the $3+1$-dimensional Dirac representation, Clifford multiplication by covectors in the restricted bundle

$$i^*T^*N \longrightarrow M$$

is defined fiberwise on S. In particular, tangent covectors $\xi \in T^*M$ and the normal covector e^0 both act on S.

The asymptotic ends play a special role. Fix one end M_ℓ, and let

$$\Phi_\ell : \mathbb{R}^3 \setminus K_\ell \longrightarrow M_\ell$$

be the diffeomorphism from Definition 1.4, where $K_\ell \subset \mathbb{R}^3$ is contractible and compact.

Lemma 2.15. *The spin structure on M restricts to a unique spin structure on each end M_ℓ. Consequently, with respect to the chosen asymptotic chart Φ_ℓ, the bundle S is trivial over M_ℓ, and there is a distinguished space of constant spinors on M_ℓ.*

Proof. Spin structures on an oriented manifold form a torsor over $H^1(\,\cdot\,;\mathbb{Z}_2)$. It is therefore enough to show that

$$H^1(\mathbb{R}^3 \setminus K_\ell;\mathbb{Z}_2) = 0.$$

Because K_ℓ is compact and contractible, Alexander duality gives

$$\widetilde{H}^1(\mathbb{R}^3 \setminus K_\ell;\mathbb{Z}_2) \cong \widetilde{H}_1(K_\ell;\mathbb{Z}_2) = 0.$$

Hence $H^1(\mathbb{R}^3 \setminus K_\ell;\mathbb{Z}_2) = 0$, so the induced spin structure on the end is unique.

Now $\mathbb{R}^3 \setminus K_\ell$ is an open subset of $\mathbb{R}^3$, and its tangent bundle is therefore trivial. Since the spin structure on this trivial oriented bundle is unique, the induced spin bundle is the trivial principal Spin(3)-bundle. Passing to the associated vector bundle shows that S is trivial over the end. Pulling back constant sections of the trivial bundle

$$(\mathbb{R}^3 \setminus K_\ell) \times S_{3,1}$$

by Φ_ℓ^{-1} gives the distinguished constant spinors on M_ℓ. □

These asymptotically constant spinors are the boundary data that will later be prescribed in Witten's equation.

2.4. **The ambient and intrinsic spin connections.** The bundle $S \to M$ carries two different natural connections.

The first comes from the spacetime Levi-Civita connection ∇^N. Restricting ∇^N to the bundle $i^*TN \to M$ gives an $SO(3,1)$-connection on the oriented orthonormal frame bundle of i^*TN, hence a lifted $\mathrm{Spin}(3,1)$-connection on the associated spinor bundle S. We denote the resulting covariant derivative by V.

The second comes from the Riemannian Levi-Civita connection ∇ of (M,g). This gives an $SO(3)$-connection on $F_{\mathrm{SO}}(M)$, hence a $\mathrm{Spin}(3)$-connection on $P_{\mathrm{Spin}}(M)$, and therefore a covariant derivative on the same vector bundle S. By a slight abuse of notation we again denote this induced spin connection by ∇.

Thus the same underlying bundle S carries two geometrically different connections:

- V, which remembers the ambient Lorentzian geometry of N,
- ∇, which remembers only the intrinsic Riemannian geometry of M.

In local orthonormal frames the two connections have the usual spinorial form. Let

$$\{e^0, e^1, e^2, e^3\}$$

be a local orthonormal coframe of i^*T^*N along M, adapted so that e^1, e^2, e^3 are tangent to M. Write ω_{ab} for the connection 1-forms of ∇^N:

$$\nabla^N e_a = \omega_{ab}\, e_b, \qquad \omega_{ab} + \omega_{ba} = 0.$$

Then for $X \in TM$ and $\psi \in \Gamma(S)$,

$$V_X \psi = X(\psi) + \frac{1}{4}\,\omega_{ab}(X)\, e^a \cdot e^b \cdot \psi. \tag{2.16}$$

Likewise, if θ_{ij} are the connection 1-forms of the Levi-Civita connection of (M,g),

$$\nabla e_i = \theta_{ij}\, e_j, \qquad \theta_{ij} + \theta_{ji} = 0,$$

then the intrinsic spin connection is

$$\nabla_X \psi = X(\psi) + \frac{1}{4}\,\theta_{ij}(X)\, e^i \cdot e^j \cdot \psi. \tag{2.17}$$

The formulas (2.16) and (2.17) imply the usual Leibniz rules with Clifford multiplication. If x is a section of i^*T^*N, then

$$V_X(x \cdot \psi) = (\nabla^N_X x) \cdot \psi + x \cdot V_X \psi. \tag{2.18}$$

If ξ is tangent, that is $\xi \in \Gamma(T^*M)$, then

$$\nabla_X(\xi \cdot \psi) = (\nabla_X \xi) \cdot \psi + \xi \cdot \nabla_X \psi. \tag{2.19}$$

These are immediate from the fact that the spin representation differentiates the orthogonal group action.

The two Hermitian structures discussed above interact differently with these two connections.

Proposition 2.20. *The ambient connection V is compatible with the indefinite Hermitian form $(\,\cdot\,,\,\cdot\,)$, while the intrinsic connection ∇ is compatible with the positive definite Hermitian inner product $\langle\,\cdot\,,\,\cdot\,\rangle$. In particular, for $X \in TM$,*

$$X(\psi,\phi) = (V_X\psi,\phi) + (\psi, V_X\phi), \tag{2.21}$$

and

$$X\langle\psi,\phi\rangle = \langle\nabla_X\psi,\phi\rangle + \langle\psi,\nabla_X\phi\rangle. \tag{2.22}$$

Proof. Equation (2.21) follows because V is induced from the spacetime $\mathrm{Spin}(3,1)$-connection, and the form $(\,\cdot\,,\,\cdot\,)$ is $\mathrm{Spin}(3,1)$-invariant.

For (2.22), recall that $\langle\psi,\phi\rangle = (e^0\cdot\psi,\phi)$. Since the intrinsic $\mathrm{Spin}(3)$-connection fixes the distinguished timelike covector e^0, we have

$$\nabla_X(e^0\cdot\psi) = e^0\cdot\nabla_X\psi.$$

Using the $\mathrm{Spin}(3)$-invariance of $(\,\cdot\,,\,\cdot\,)$, we obtain

$$X\langle\psi,\phi\rangle = X(e^0\cdot\psi,\phi) = (e^0\cdot\nabla_X\psi,\phi) + (e^0\cdot\psi,\nabla_X\phi) = \langle\nabla_X\psi,\phi\rangle + \langle\psi,\nabla_X\phi\rangle.$$

□

The distinction between V and ∇ is the geometric heart of the spinorial proof. The hypersurface Dirac operator is defined using V, not ∇, and the difference between the two turns out to be measured exactly by the second fundamental form h_{ij}. We turn to this in the next section.

3. The hypersurface Dirac operator

We now introduce the Dirac operator that enters Witten's argument. The underlying bundle is the Dirac spinor bundle $S \to M$ from Section 2, but the connection used to define the operator is the ambient spin connection V, not the intrinsic Riemannian spin connection ∇. This is exactly the point at which the extrinsic geometry of the hypersurface enters the proof.

3.1. Definition of the hypersurface Dirac operator.

Let $\{e_1,e_2,e_3\}$ be a local oriented g-orthonormal frame of TM, and let $\{e^1,e^2,e^3\}$ be the dual coframe. Recall from Section 2 that the ambient spin connection is a covariant derivative

$$V : \Gamma(S) \longrightarrow \Gamma(T^*M\otimes S).$$

Clifford multiplication by tangent covectors gives a bundle map

$$c : T^*M\otimes S \longrightarrow S.$$

Definition 3.1. The *hypersurface Dirac operator* is the first-order differential operator

$$\mathscr{D} : \Gamma(S) \longrightarrow \Gamma(S)$$

defined by

$$\mathscr{D}\psi = \sum_{i=1}^{3} e^i\cdot V_i\psi, \qquad V_i := V_{e_i}. \tag{3.2}$$

The formula (3.2) is independent of the choice of local oriented orthonormal frame. Indeed, if $\{\widetilde{e}_i\}$ is another such frame, then $\widetilde{e}_i = A_i{}^j e_j$ for some $SO(3)$-valued matrix A, and $\widetilde{e}^i = A^i{}_j e^j$. Hence

$$\sum_i \widetilde{e}^i \cdot V_{\widetilde{e}_i}\psi = \sum_{i,j,k} A^i{}_j A_i{}^k\, e^j \cdot V_k\psi = \sum_j e^j \cdot V_j\psi.$$

For comparison, the intrinsic Dirac operator of the Riemannian manifold (M, g) is

$$\nabla\!\!\!/\psi = \sum_{i=1}^{3} e^i \cdot \nabla_i\psi.$$

The difference between $\mathscr{D}$ and $\nabla\!\!\!/$ is therefore exactly the difference between the two spin connections V and ∇.

Proposition 3.3. *The operators $\mathscr{D}$ and $\nabla\!\!\!/$ have the same principal symbol. More precisely, if $\xi = \xi_i e^i \in T^*_x M$, then*

$$\sigma_\xi(\mathscr{D}) = \sigma_\xi(\nabla\!\!\!/) = \xi \cdot . \tag{3.4}$$

In particular, $\mathscr{D}$ is elliptic.

Proof. The principal symbol of a first-order operator depends only on its first-derivative part. Since $V_i - \nabla_i$ is a bundle endomorphism, the two operators differ by a zero-order term and therefore have the same principal symbol. Explicitly,

$$\sigma_\xi(\mathscr{D})\psi = \sum_{i=1}^{3} \xi_i\, e^i \cdot \psi = \xi \cdot \psi.$$

By the Clifford relations,

$$(\xi\cdot)^2 = -|\xi|^2 \mathrm{Id}_S$$

for every nonzero $\xi \in T^*_x M$, so $\xi\cdot$ is invertible whenever $\xi \neq 0$. Hence $\mathscr{D}$ is elliptic. □

3.2. **Relation between V and ∇.** We now compute the difference between the ambient and intrinsic spin connections. This is the place where the second fundamental form enters.

Let $\{e^0, e^1, e^2, e^3\}$ be a local orthonormal coframe of i^*T^*N along M, adapted so that e^1, e^2, e^3 are tangent to M. Let $\omega_{\alpha\beta}$ be the connection 1-forms of the spacetime Levi-Civita connection:

$$\nabla^N e_\alpha = \omega_{\alpha\beta}\, e_\beta, \qquad \omega_{\alpha\beta} + \omega_{\beta\alpha} = 0.$$

Let θ_{ij} be the connection 1-forms of the Levi-Civita connection of (M, g):

$$\nabla e_i = \theta_{ij}\, e_j, \qquad \theta_{ij} + \theta_{ji} = 0.$$

The second fundamental form was defined in Section 1 by

$$h_{ij} = g(\nabla^N_{e_i} e_0, e_j).$$

Equivalently,

$$\omega_{0j}(e_i) = h_{ij}, \qquad \omega_{j0}(e_i) = -h_{ij}. \tag{3.5}$$

Also, because the tangential part of ∇^N along M is the Levi-Civita connection of (M, g), we have

$$\omega_{ij}|_{TM} = \theta_{ij}. \tag{3.6}$$

By Section 2, the two spin connections are

$$V_X\psi = X(\psi) + \frac{1}{4}\,\omega_{\alpha\beta}(X)\,e^{\alpha}\cdot e^{\beta}\cdot\psi, \tag{3.7}$$

and

$$\nabla_X\psi = X(\psi) + \frac{1}{4}\,\theta_{ij}(X)\,e^{i}\cdot e^{j}\cdot\psi. \tag{3.8}$$

Proposition 3.9. *For every tangent frame vector e_i,*

$$V_i\psi = \nabla_i\psi + \frac{1}{2}\,h_{ij}\,e^0\cdot e^j\cdot\psi = \nabla_i\psi - \frac{1}{2}\,h_{ij}\,e^j\cdot e^0\cdot\psi. \tag{3.10}$$

Proof. Set $X = e_i$ in (3.7). Using (3.6), (3.5), and $\omega_{j0} = -\omega_{0j}$, we obtain

$$\begin{aligned} V_i\psi &= e_i(\psi) + \frac{1}{4}\,\theta_{jk}(e_i)\,e^j\cdot e^k\cdot\psi + \frac{1}{4}\,\omega_{0j}(e_i)\,e^0\cdot e^j\cdot\psi + \frac{1}{4}\,\omega_{j0}(e_i)\,e^j\cdot e^0\cdot\psi \\ &= \nabla_i\psi + \frac{1}{4}\,h_{ij}\,e^0\cdot e^j\cdot\psi - \frac{1}{4}\,h_{ij}\,e^j\cdot e^0\cdot\psi. \end{aligned}$$

Since $e^0\cdot e^j = -e^j\cdot e^0$, the last two terms combine to

$$\frac{1}{4}\,h_{ij}\big(e^0\cdot e^j - e^j\cdot e^0\big)\psi = \frac{1}{2}\,h_{ij}\,e^0\cdot e^j\cdot\psi.$$

This gives the first equality in (3.10); the second is just the anticommutation relation between e^0 and e^j. □

Two immediate consequences are worth recording.

Corollary 3.11. *The difference $V - \nabla$ is a zero-order operator:*

$$(V_i - \nabla_i)\psi = \frac{1}{2}\,h_{ij}\,e^0\cdot e^j\cdot\psi.$$

Consequently,

$$\mathscr{D} - \not{\nabla} = \sum_{i=1}^{3} e^i\cdot(V_i - \nabla_i)$$

is also a zero-order operator.

Corollary 3.12. *In any local orthonormal frame adapted to M,*

$$V_i e^j = -h_{ij}e^0, \qquad V_i e^0 = -h_{ij}e^j. \tag{3.13}$$

Proof. The second identity follows from the definition of the second fundamental form, since

$$\nabla^N_{e_i} e_0 = h_{ij}e_j.$$

Passing to the dual coframe introduces a minus sign, so

$$V_i e^0 = -h_{ij}e^j.$$

Similarly, $\nabla^N_{e_i} e_j$ has normal component $-h_{ij}e_0$, hence the covector e^j varies by

$$V_i e^j = -h_{ij}e^0.$$

□

3.3. **Formal adjoints.** From now on we write $\mu = e^1 \wedge e^2 \wedge e^3$ for the Riemannian volume form of (M, g), and we use the positive definite Hermitian inner product $\langle \cdot\,, \cdot \rangle$ from Section 2. Recall that

$$\langle \psi, \phi \rangle = (e^0 \cdot \psi, \phi),$$

where $(\,\cdot\,,\,\cdot\,)$ is the indefinite Hermitian form preserved by the ambient connection V.

Because V preserves $(\,\cdot\,,\,\cdot\,)$ but not $\langle \cdot\,, \cdot \rangle$, the formal adjoint of V contains an extra zero-order term coming from the variation of e^0.

Lemma 3.14. *For $\phi, \psi \in \Gamma(S)$ and $1 \le i \le 3$,*

$$e_i \langle \phi, \psi \rangle = \langle V_i \phi, \psi \rangle + \langle \phi, V_i \psi \rangle + \langle h_{ij} e^j \cdot e^0 \cdot \phi, \psi \rangle. \tag{3.15}$$

Proof. Using the definition of $\langle \cdot\,, \cdot \rangle$ and the fact that V is compatible with $(\,\cdot\,,\,\cdot\,)$, we compute

$$\begin{aligned} e_i \langle \phi, \psi \rangle &= e_i \left(e^0 \cdot \phi, \psi \right) \\ &= \big((V_i e^0) \cdot \phi, \psi \big) + \big(e^0 \cdot V_i \phi, \psi \big) + \big(e^0 \cdot \phi, V_i \psi \big). \end{aligned}$$

By (3.13),

$$V_i e^0 = -h_{ij} e^j,$$

so the first term becomes

$$\big((V_i e^0) \cdot \phi, \psi \big) = \big(-h_{ij} e^j \cdot \phi, \psi \big) = \langle h_{ij} e^j \cdot e^0 \cdot \phi, \psi \rangle.$$

The remaining two terms are $\langle V_i \phi, \psi \rangle$ and $\langle \phi, V_i \psi \rangle$, and (3.15) follows. □

We can now compute the formal adjoint of the connection.

Proposition 3.16. *For each i, the formal adjoint of V_i with respect to the L^2-inner product induced by $\langle \cdot\,, \cdot \rangle$ is*

$$V_i^* = -\,V_i - h_{ij} e^j \cdot e^0 \cdot\,. \tag{3.17}$$

Equivalently, for compactly supported $\phi, \psi \in \Gamma(S)$,

$$\int_M \langle \phi, V_i \psi \rangle \, \mu = -\int_M \langle V_i \phi + h_{ij} e^j \cdot e^0 \cdot \phi, \psi \rangle \, \mu. \tag{3.18}$$

Proof. Choose a local orthonormal frame near a point $p \in M$ such that

$$(\nabla_{e_i} e_j)_p = 0.$$

Then at p,

$$d\big(\langle \phi, \psi \rangle \, e_i \lrcorner \mu \big) = e_i \langle \phi, \psi \rangle \, \mu.$$

Applying Lemma 3.14 gives

$$d\big(\langle \phi, \psi \rangle \, e_i \lrcorner \mu \big) = \big(\langle V_i \phi, \psi \rangle + \langle \phi, V_i \psi \rangle + \langle h_{ij} e^j \cdot e^0 \cdot \phi, \psi \rangle \big) \mu.$$

Integrating over M and using Stokes' theorem for compactly supported spinors proves (3.18), hence (3.17). □

Next we turn to the hypersurface Dirac operator.

Proposition 3.19. *The operator $\mathscr{D}$ is formally self-adjoint:*

$$\mathscr{D}^* = \mathscr{D}. \tag{3.20}$$

More explicitly, for compactly supported $\phi, \psi \in \Gamma(S)$,

$$\int_M \langle \phi, \mathscr{D}\psi \rangle \, \mu = \int_M \langle \mathscr{D}\phi, \psi \rangle \, \mu. \tag{3.21}$$

Proof. At a point p where $(\nabla_{e_i} e_j)_p = 0$, set

$$\alpha_i := \langle \phi, e^i \cdot \psi \rangle.$$

Since $e^i\cdot$ is skew-Hermitian with respect to $\langle \cdot\,, \cdot \rangle$, we have

$$\alpha_i = -\langle e^i \cdot \phi, \psi \rangle.$$

Using $d(\alpha_i \, e_i \lrcorner \mu) = e_i(\alpha_i)\mu$ at p, the Leibniz rule (2.18), and the skew-Hermitian property of Clifford multiplication by tangent covectors, we compute

$$\begin{aligned} d\big(\langle \phi, e^i \cdot \psi \rangle \, e_i \lrcorner \mu\big) &= e_i \langle \phi, e^i \cdot \psi \rangle \, \mu \\ &= \big(\langle V_i \phi, e^i \cdot \psi \rangle + \langle \phi, (V_i e^i) \cdot \psi \rangle + \langle \phi, e^i \cdot V_i \psi \rangle + \langle h_{ij} e^j \cdot e^0 \cdot \phi, e^i \cdot \psi \rangle \big) \mu. \end{aligned}$$

At p, we have $V_i e^i = -h_{ii} e^0$ by (3.13). The second and fourth terms cancel after moving $e^i\cdot$ across the inner product and using the Clifford relations together with the symmetry of h_{ij}. The remaining terms give

$$d\big(\langle \phi, e^i \cdot \psi \rangle \, e_i \lrcorner \mu\big) = \big(\langle \phi, \mathscr{D}\psi \rangle - \langle \mathscr{D}\phi, \psi \rangle \big) \mu.$$

Integrating and applying Stokes' theorem for compactly supported spinors yields (3.21). □

It is convenient to package the componentwise adjoints into the covariant Laplacian determined by V:

$$V^* V := \sum_{i=1}^{3} V_i^* V_i. \tag{3.22}$$

By (3.17),

$$V^* V = -\sum_{i=1}^{3} V_i V_i - \sum_{i,j=1}^{3} h_{ij} e^j \cdot e^0 \cdot V_i. \tag{3.23}$$

3.4. **The Weitzenböck formula.** We now compute the square of $\mathscr{D}$. This is the key local identity in the spinorial proof.

Throughout this subsection, $R_{\alpha\beta\gamma\delta}$, $R_{\alpha\beta}$, and R denote the spacetime Riemann tensor, Ricci tensor, and scalar curvature of (N, g), all restricted to the hypersurface M.

Theorem 3.24 (Weitzenböck formula)**.** *The hypersurface Dirac operator satisfies*

$$\mathscr{D}^2 = V^* V + \mathcal{R}, \tag{3.25}$$

where

$$\mathcal{R} = \frac{1}{4}\Big(R + 2R_{00} + 2R_{0i} \, e^0 \cdot e^i \cdot \Big). \tag{3.26}$$

Proof. Fix $p \in M$. Choose an orthonormal basis $\{e_0, e_1, e_2, e_3\}$ of T_pN with e_0 normal and e_1, e_2, e_3 tangent to M. Extend e_1, e_2, e_3 to a local orthonormal frame on M so that

$$(\nabla_{e_i} e_j)_p = 0 \qquad (1 \le i, j \le 3).$$

Then extend further to an orthonormal frame $\{e_0, e_1, e_2, e_3\}$ on N so that

$$(\nabla^N_{e_0} e_i)_p = 0 \qquad (1 \le i \le 3).$$

Let $\{e^0, e^1, e^2, e^3\}$ be the dual coframe. By Corollary 3.12,

$$(V_i e^j)_p = -h_{ij} e^0, \qquad (V_i e^0)_p = -h_{ij} e^j. \tag{3.27}$$

Also, because the tangential frame is geodesic on M at p,

$$[e_i, e_j]_p = 0.$$

Starting from (3.2), we compute at p:

$$\begin{aligned} \mathscr{D}^2\psi &= \sum_{i,j=1}^{3} e^i \cdot V_i\big(e^j \cdot V_j\psi\big) \\ &= \sum_{i,j=1}^{3} e^i \cdot e^j \cdot V_i V_j \psi + \sum_{i,j=1}^{3} e^i \cdot (V_i e^j) \cdot V_j \psi. \end{aligned} \tag{3.28}$$

Using (3.27), the second term becomes

$$\sum_{i,j=1}^{3} e^i \cdot (V_i e^j) \cdot V_j \psi = -\sum_{i,j=1}^{3} h_{ij}\, e^i \cdot e^0 \cdot V_j \psi. \tag{3.29}$$

We now treat the first term in (3.28). Split it into diagonal and off-diagonal parts:

$$\sum_{i,j} e^i \cdot e^j \cdot V_i V_j = \sum_i e^i \cdot e^i \cdot V_i V_i + \sum_{i \ne j} e^i \cdot e^j \cdot V_i V_j.$$

Because $e^i \cdot e^i = -1$, the diagonal part is

$$\sum_i e^i \cdot e^i \cdot V_i V_i = -\sum_i V_i V_i. \tag{3.30}$$

For the off-diagonal part, antisymmetry of Clifford multiplication gives

$$\begin{aligned} \sum_{i \ne j} e^i \cdot e^j \cdot V_i V_j &= \frac{1}{2} \sum_{i \ne j} e^i \cdot e^j \cdot (V_i V_j - V_j V_i) \\ &= \sum_{i<j} e^i \cdot e^j \cdot \Omega_{ij}, \end{aligned} \tag{3.31}$$

where

$$\Omega_{ij} := [V_i, V_j]$$

is the curvature of the ambient spin connection.

Let

$$p : \mathfrak{so}(3,1) \longrightarrow \mathrm{End}(S)$$

be the infinitesimal spin representation. With our Clifford conventions,

$$p(e^\alpha \wedge e^\beta) = -\frac{1}{2}\, e^\alpha \cdot e^\beta. \tag{3.32}$$

Hence

$$\Omega_{ij} = R_{ij\alpha\beta}\, p(e^\alpha \wedge e^\beta) = -\frac{1}{2}\, R_{ij\alpha\beta}\, e^\alpha \cdot e^\beta, \tag{3.33}$$

and therefore

$$\sum_{i<j} e^i \cdot e^j \cdot \Omega_{ij} = -\frac{1}{2} \sum_{i<j} R_{ij\alpha\beta}\, e^i \cdot e^j \cdot e^\alpha \cdot e^\beta. \tag{3.34}$$

We now simplify (3.34). There are two types of terms.

First, the purely spatial terms $(\alpha, \beta) = (k, \ell)$ contribute a scalar. Using the symmetries of the Riemann tensor and the Clifford relations, one finds

$$-\frac{1}{2} \sum_{i<j,\, k<\ell} R_{ijk\ell}\, e^i \cdot e^j \cdot e^k \cdot e^\ell = \frac{1}{4}\,(R + 2R_{00})\, \mathrm{Id}_S.$$

Second, the mixed terms $(\alpha, \beta) = (0, k)$ contribute the Clifford-linear term

$$-\frac{1}{2} \sum_{i<j,\, k} R_{ij0k}\, e^i \cdot e^j \cdot e^0 \cdot e^k = \frac{1}{2}\, R_{0k}\, e^0 \cdot e^k.$$

The first Bianchi identity and the curvature symmetries are exactly what reduce these sums to Ricci contractions. Altogether,

$$\sum_{i<j} e^i \cdot e^j \cdot \Omega_{ij} = \frac{1}{4}\,(R + 2R_{00}) + \frac{1}{2}\, R_{0i}\, e^0 \cdot e^i. \tag{3.35}$$

Combining (3.28), (3.29), (3.30), and (3.35), we obtain

$$\mathscr{D}^2 = -\sum_i V_i V_i - \sum_{i,j} h_{ij} e^i \cdot e^0 \cdot V_j + \frac{1}{4}\,(R + 2R_{00}) + \frac{1}{2}\, R_{0i} e^0 \cdot e^i. \tag{3.36}$$

Finally, compare this with (3.23). Since $h_{ij} = h_{ji}$, we may relabel the dummy indices in (3.23) to write

$$V^* V = -\sum_i V_i V_i - \sum_{i,j} h_{ij} e^i \cdot e^0 \cdot V_j.$$

Substituting this into (3.36) gives

$$\mathscr{D}^2 = V^* V + \frac{1}{4}\,(R + 2R_{00}) + \frac{1}{2}\, R_{0i} e^0 \cdot e^i,$$

which is exactly (3.25)–(3.26). □

The curvature term in Theorem 3.24 is precisely the one needed in the positive energy theorem, because Einstein's equation turns it into an expression controlled by the dominant energy condition.

Corollary 3.37. *Assume that the spacetime satisfies Einstein's equation*

$$R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R = 8\pi T_{\alpha\beta}.$$

Then the endomorphism $\mathcal{R}$ in (3.26) can be written as

$$\mathcal{R} = 4\pi \Big(T_{00} + T_{0i}\, e^0 \cdot e^i \cdot \Big). \tag{3.38}$$

Proof. Because $g_{00} = -1$ and $g_{0i} = 0$ in an adapted orthonormal frame, Einstein's equation gives

$$R_{00} + \frac{1}{2}R = 8\pi T_{00}, \qquad R_{0i} = 8\pi T_{0i}.$$

Substituting these identities into (3.26) yields

$$\mathcal{R} = \frac{1}{4}\Big((R + 2R_{00}) + 2R_{0i}e^0 \cdot e^i\Big) = 4\pi T_{00} + 4\pi T_{0i}e^0 \cdot e^i,$$

which is (3.38). □

3.5. **Positivity of the zeroth-order term.** We now show that the dominant energy condition makes the zeroth-order term in the Weitzenböck formula nonnegative.

Proposition 3.39. *Assume the dominant energy condition along M. Then the endomorphism $\mathcal{R}$ is pointwise nonnegative in the sense that*

$$\langle \psi, \mathcal{R}\psi \rangle \geq 0 \qquad \textit{for every } \psi \in S. \tag{3.40}$$

Proof. By Corollary 3.37, it is enough to study the Hermitian endomorphism

$$A := T_{00} + T_{0i}\, e^0 \cdot e^i.$$

Set

$$B := T_{0i}\, e^0 \cdot e^i.$$

By Lemma 2.12, each operator $e^0 \cdot e^i$ is Hermitian with respect to $\langle \cdot, \cdot \rangle$, so B is Hermitian.

We first compute B^2. Using the Clifford relations and $(e^0)^2 = 1$, we have

$$e^0 \cdot e^i \cdot e^0 = -\, e^i,$$

hence

$$e^0 \cdot e^i \cdot e^0 \cdot e^j = -\, e^i \cdot e^j.$$

Therefore

$$B^2 = -\, T_{0i}T_{0j}\, e^i \cdot e^j.$$

Split the sum into diagonal and off-diagonal parts. Since $e^i \cdot e^i = -1$, the diagonal part gives

$$-\sum_i T_{0i}^2\, e^i \cdot e^i = \sum_i T_{0i}^2\, \mathrm{Id}_S.$$

For $i \neq j$, the terms cancel in pairs because

$$e^i \cdot e^j + e^j \cdot e^i = 0.$$

Thus

$$B^2 = \Big(\sum_{i=1}^{3} T_{0i}^2\Big)\mathrm{Id}_S. \tag{3.41}$$

Since B is Hermitian and satisfies (3.41), its eigenvalues are

$$\pm\Big(\sum_{i=1}^{3} T_{0i}^2\Big)^{1/2}.$$

Consequently, the eigenvalues of

$$A = T_{00}\mathrm{Id}_S + B$$

are

$$T_{00} \pm \Big(\sum_{i=1}^{3} T_{0i}^2\Big)^{1/2}.$$

By the dominant energy condition,

$$T_{00} \geq \Big(\sum_{i=1}^{3} T_{0i}^2\Big)^{1/2},$$

so both eigenvalues are nonnegative. Hence A is a nonnegative Hermitian endomorphism, and therefore

$$\langle \psi, A\psi \rangle \geq 0 \qquad \text{for all } \psi \in S.$$

Multiplying by 4π and using (3.38) gives (3.40). □

The content of this section may be summarized as follows: the square of the hypersurface Dirac operator is the covariant Laplacian V^*V plus a zeroth-order term $\mathcal{R}$, and under the dominant energy condition that zeroth-order term is nonnegative. In the next section, after integrating by parts, the only remaining ingredient will be the boundary term at infinity.

4. The integral identity and the boundary term at infinity

In Section 3 we obtained the Weitzenböck formula

$$\mathscr{D}^2 = V^*V + \mathcal{R}$$

for the hypersurface Dirac operator. We now pass from this local identity to an integral identity and then compute the boundary contribution at infinity. The boundary term is where the ADM energy-momentum enters.

4.1. The integrated Weitzenböck identity.

Let $\Omega \subset M$ be a compact domain with smooth boundary. We continue to write

$$\mu = e^1 \wedge e^2 \wedge e^3$$

for the Riemannian volume form on M, and we use the positive definite Hermitian inner product $\langle \cdot, \cdot \rangle$ on the spinor bundle S.

The boundary term in the integral formula is most naturally written as a 2-form.

Definition 4.1. For a smooth spinor $\psi \in \Gamma(S)$, define the 2-form

$$\beta(\psi) := -\frac{1}{2} \left\langle \psi,\, [e^i, e^j] \cdot V_j \psi \right\rangle (e_i \lrcorner \mu), \tag{4.2}$$

where

$$[e^i, e^j] := e^i \cdot e^j - e^j \cdot e^i$$

is the Clifford commutator.

The reason for this choice is the algebraic identity

$$V_i \psi + e^i \cdot \mathscr{D}\psi = \frac{1}{2} [e^i, e^j] \cdot V_j \psi. \tag{4.3}$$

Indeed,

$$e^i \cdot \mathscr{D}\psi = e^i \cdot e^j \cdot V_j \psi,$$

so

$$V_i \psi + e^i \cdot \mathscr{D}\psi = (\delta_{ij} + e^i \cdot e^j)\, V_j \psi.$$

For tangent covectors we have

$$e^i \cdot e^j + e^j \cdot e^i = -2\delta_{ij},$$

hence

$$[e^i, e^j] = e^i \cdot e^j - e^j \cdot e^i = 2(\delta_{ij} + e^i \cdot e^j),$$

which gives (4.3).

We next derive the integrated form of the Weitzenböck identity.

Proposition 4.4. *For every smooth spinor ψ on a compact domain $\Omega \subset M$ with smooth boundary,*

$$\int_\Omega \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi \rangle - |\mathscr{D}\psi|^2 \Big)\, \mu = \int_{\partial\Omega} \beta(\psi). \tag{4.5}$$

Proof. Starting from the Weitzenböck formula,

$$\mathscr{D}^2\psi = V^*V\psi + \mathcal{R}\psi,$$

take the inner product with ψ:

$$\langle \psi, \mathscr{D}^2\psi \rangle = \langle \psi, V^*V\psi \rangle + \langle \psi, \mathcal{R}\psi \rangle. \tag{4.6}$$

We first rewrite the term involving V^*V. In the local orthonormal frame used in Section 3, the integration-by-parts formula for V was

$$d\big(\langle \phi, V_i\chi \rangle\, e_i \lrcorner \mu\big) = \big(\langle V_i\phi, V_i\chi \rangle + \langle \phi, V_i^* V_i \chi \rangle\big)\mu.$$

Set $\phi = \chi = \psi$ and sum over i. Since

$$V^*V = \sum_i V_i^* V_i, \qquad |V\psi|^2 = \sum_i \langle V_i\psi, V_i\psi \rangle,$$

we obtain

$$\langle \psi, V^*V\psi \rangle\, \mu = d\big(\langle \psi, V_i\psi \rangle\, e_i \lrcorner \mu\big) - |V\psi|^2\, \mu. \tag{4.7}$$

Next we do the same for $\mathscr{D}$. The formal self-adjointness computation in Section 3 gives

$$d\big(\langle \phi, e^i \cdot \chi \rangle\, e_i \lrcorner \mu\big) = \big(\langle \phi, \mathscr{D}\chi \rangle - \langle \mathscr{D}\phi, \chi \rangle\big)\mu.$$

Taking $\phi = \psi$ and $\chi = \mathscr{D}\psi$, and using $\mathscr{D}^* = \mathscr{D}$, we get

$$\langle \psi, \mathscr{D}^2\psi \rangle\, \mu = d\big(\langle \psi, e^i \cdot \mathscr{D}\psi \rangle\, e_i \lrcorner \mu\big) + |\mathscr{D}\psi|^2\, \mu. \tag{4.8}$$

Substitute (4.7) and (4.8) into (4.6). This gives

$$d\big(\langle \psi, e^i \cdot \mathscr{D}\psi \rangle\, e_i \lrcorner \mu\big) + |\mathscr{D}\psi|^2\, \mu = d\big(\langle \psi, V_i\psi \rangle\, e_i \lrcorner \mu\big) - |V\psi|^2\, \mu + \langle \psi, \mathcal{R}\psi \rangle\, \mu.$$

Rearranging,

$$\Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi \rangle - |\mathscr{D}\psi|^2 \Big)\mu = d\Big(\langle \psi, e^i \cdot \mathscr{D}\psi \rangle\, e_i \lrcorner \mu - \langle \psi, V_i\psi \rangle\, e_i \lrcorner \mu \Big).$$

By (4.3), the 2-form inside the differential is exactly $\beta(\psi)$. Hence

$$\Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi \rangle - |\mathscr{D}\psi|^2 \Big)\mu = d\beta(\psi).$$

Integrating over Ω and applying Stokes' theorem proves (4.5). □

4.2. **The case $\mathscr{D}\psi = 0$.** The integral formula simplifies immediately for harmonic spinors of the hypersurface Dirac operator.

Corollary 4.9. *If ψ satisfies*

$$\mathscr{D}\psi = 0$$

on Ω, then

$$\int_\Omega \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi \rangle \Big)\, \mu = \int_{\partial\Omega} \beta(\psi). \tag{4.10}$$

Proof. This is (4.5) with $|\mathscr{D}\psi|^2 = 0$. □

Under the dominant energy condition, the zeroth-order term is nonnegative by Proposition 3.39. Therefore the left-hand side of (4.10) is nonnegative.

We will later apply (4.10) to a truncation of M. For R large, let

$$M_R := M \setminus \bigcup_{\ell=1}^{k} M_{\ell,R},$$

so that

$$\partial M_R = \bigsqcup_{\ell=1}^{k} S_{R,\ell},$$

where $S_{R,\ell}$ is the Euclidean coordinate sphere of radius R in the ℓ-th end. Then, for any harmonic spinor ψ,

$$\int_{M_R} \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi \rangle \Big)\, \mu = \sum_{\ell=1}^{k} \int_{S_{R,\ell}} \beta(\psi). \tag{4.11}$$

The remaining task in this section is to identify the limit of the right-hand side as $R \to \infty$.

4.3. **Asymptotic expansion of the boundary integrand.** Fix one asymptotically flat end M_ℓ, and identify it with the corresponding subset of $\mathbb{R}^3$ by the asymptotic chart. Let

$$x = (x^1, x^2, x^3), \qquad r = |x|, \qquad S_r = \{\, |x| = r \,\}.$$

We write

$$d\Omega^i := \partial_i \lrcorner\, (dx^1 \wedge dx^2 \wedge dx^3) = \nu^i\, dS$$

for the Euclidean surface element on S_r.

By Section 2, the spinor bundle S is trivial over the end, and there is a distinguished space of constant spinors. Fix one such constant spinor and denote it by

$$\psi_\infty.$$

Choose an orthonormal coframe $\{e^1, e^2, e^3\}$ on the end satisfying

$$e^i = dx^i + O(r^{-1}). \tag{4.12}$$

The asymptotic decay of the metric implies that the connection coefficients of ∇ are $O(r^{-2})$, and therefore

$$V\psi_\infty = O(r^{-2}). \tag{4.13}$$

The boundary integral can therefore be computed using the coordinate coframe $\{dx^i\}$ in place of $\{e^i\}$.

Lemma 4.14. *For a constant spinor ψ_∞ on the end,*

$$\int_{S_r} \beta(\psi_\infty) = -\frac{1}{2}\int_{S_r} \langle \psi_\infty,\, [dx^i, dx^j]\cdot V_j\psi_\infty\rangle\, d\Omega^i + o(1) \qquad (r\to\infty).$$

Proof. By (4.12), replacing e^i by dx^i changes the Clifford factors in $\beta(\psi_\infty)$ by $O(r^{-1})$. Since $V\psi_\infty = O(r^{-2})$ by (4.13), the pointwise error in the integrand is $O(r^{-3})$. Integrating over S_r, whose area is $O(r^2)$, produces an $O(r^{-1})$ error, which tends to 0. □

We next compute the leading term of $V_j\psi_\infty$.

Let $\Gamma_{kj\ell}$ denote the Christoffel symbols of the metric g on the end, with the first index lowered using the Euclidean tensor δ_{ij}:

$$\Gamma_{kj\ell} = \frac{1}{2}\big(\partial_j g_{k\ell} + \partial_\ell g_{kj} - \partial_k g_{j\ell}\big).$$

Because $g_{ij} = \delta_{ij} + O(r^{-1})$, these satisfy $\Gamma_{kj\ell} = O(r^{-2})$.

Lemma 4.15. *For a constant spinor ψ_∞ on the end,*

$$V_j\psi_\infty = -\frac{1}{4}\,\Gamma_{kj\ell}\, dx^k\cdot dx^\ell\cdot\psi_\infty + \frac{1}{2}\,h_{jk}\, dx^0\cdot dx^k\cdot\psi_\infty + O(r^{-3})\,\psi_\infty. \tag{4.16}$$

Here dx^0 denotes the constant timelike covector in the asymptotic Minkowski frame used to trivialize S over the end.

Proof. Since ψ_∞ is constant in the asymptotic trivialization, its ordinary coordinate derivatives vanish. Thus only the spin connection coefficients contribute to $V_j\psi_\infty$.

The intrinsic spin connection contributes the term involving the Christoffel symbols. More precisely, comparing the orthonormal coframe $\{e^i\}$ with the coordinate coframe $\{dx^i\}$ and using $e^i = dx^i + O(r^{-1})$, one finds

$$\nabla_j\psi_\infty = -\frac{1}{4}\,\Gamma_{kj\ell}\, dx^k\cdot dx^\ell\cdot\psi_\infty + O(r^{-3})\,\psi_\infty.$$

The ambient and intrinsic spin connections differ by

$$V_j\psi = \nabla_j\psi + \frac{1}{2}h_{jk}\, dx^0\cdot dx^k\cdot\psi$$

from Proposition 3.9, and (4.16) follows. □

We can now expand the boundary integrand explicitly.

Proposition 4.17. *For a constant spinor ψ_∞ on the end,*

$$\begin{aligned}\beta(\psi_\infty) = \frac{1}{4}\Big\langle \psi_\infty,\, \Big((\partial_j g_{ij} - \partial_i g_{jj})\mathrm{Id}_S + 2(h_{ik} - \delta_{ik}h_{jj})\, dx^0\cdot dx^k\Big)\psi_\infty\Big\rangle\, d\Omega^i \\ + O(r^{-3})\, dS.\end{aligned} \tag{4.18}$$

Proof. By Lemma 4.14 and Lemma 4.15,

$$\begin{aligned}\int_{S_r}\beta(\psi_\infty) = -\frac{1}{2}\int_{S_r}\Big\langle \psi_\infty,\, [dx^i, dx^j]\cdot\Big(-\frac{1}{4}\Gamma_{kj\ell}\, dx^k\cdot dx^\ell + \frac{1}{2}h_{jk}\, dx^0\cdot dx^k\Big)\psi_\infty\Big\rangle\, d\Omega^i \\ + o(1).\end{aligned}$$

We treat the metric term and the h-term separately.

Step 1: the term involving $\Gamma_{kj\ell}$. Using the Clifford relations, one checks that

$$[dx^i, dx^j] \cdot dx^k \cdot dx^\ell = 4(\delta^{ik}\delta^{j\ell} - \delta^{i\ell}\delta^{jk})\,\mathrm{Id}_S + 2\delta^{ik}\,dx^j \cdot dx^\ell - 2\delta^{jk}\,dx^i \cdot dx^\ell - 2\delta^{i\ell}\,dx^j \cdot dx^k + 2\delta^{j\ell}\,dx^i \cdot dx^k. \tag{4.19}$$

Because $\Gamma_{kj\ell}$ is symmetric in k and ℓ, the two-form part of (4.19) contributes zero after contraction with $\Gamma_{kj\ell}$. Thus only the scalar part remains, and the metric contribution is

$$\frac{1}{8}\,\Gamma_{kj\ell}\,4(\delta^{ik}\delta^{j\ell} - \delta^{i\ell}\delta^{jk}) = \frac{1}{2}(\Gamma_{ijj} - \Gamma_{jji}).$$

Now

$$\Gamma_{ijj} = \frac{1}{2}\big(\partial_j g_{ij} + \partial_j g_{ij} - \partial_i g_{jj}\big) = \partial_j g_{ij} - \frac{1}{2}\partial_i g_{jj},$$

while

$$\Gamma_{jji} = \frac{1}{2}\big(\partial_j g_{ji} + \partial_i g_{jj} - \partial_j g_{ji}\big) = \frac{1}{2}\partial_i g_{jj}.$$

Therefore

$$\frac{1}{2}(\Gamma_{ijj} - \Gamma_{jji}) = \frac{1}{2}(\partial_j g_{ij} - \partial_i g_{jj}).$$

Multiplying by the overall coefficient already present in the boundary integral gives the metric part of (4.18).

Step 2: the term involving h_{jk}. Again from the Clifford relations,

$$[dx^i, dx^j] \cdot dx^0 \cdot dx^k = 2\big(\delta^{jk}\,dx^0 \cdot dx^i - \delta^{ik}\,dx^0 \cdot dx^j\big). \tag{4.20}$$

Hence the h-term contributes

$$\begin{aligned} -\frac{1}{4}\,h_{jk}\,[dx^i, dx^j] \cdot dx^0 \cdot dx^k &= -\frac{1}{2}\,\big(h_{jj}\,dx^0 \cdot dx^i - h_{ik}\,dx^0 \cdot dx^k\big) \\ &= \frac{1}{2}\,(h_{ik} - \delta_{ik}h_{jj})\,dx^0 \cdot dx^k. \end{aligned}$$

Combining this with the metric term proves (4.18). □

4.4. **Identification with ADM energy-momentum.** We now convert the asymptotic expansion into a boundary limit.

Proposition 4.21. *Let* ψ_∞ *be a constant spinor on the end* M_ℓ. *Then*

$$\lim_{r\to\infty} \int_{S_r} \beta(\psi_\infty) = 4\pi\Big(E_\ell\,\langle\psi_\infty, \psi_\infty\rangle + \big\langle\psi_\infty,\, (P_\ell)_k\,dx^0 \cdot dx^k \cdot \psi_\infty\big\rangle\Big). \tag{4.22}$$

Proof. Integrate (4.18) over S_r. By the definitions of the ADM energy and momentum from Section 1,

$$E_\ell = \frac{1}{16\pi}\lim_{r\to\infty}\int_{S_r} (\partial_j g_{ij} - \partial_i g_{jj})\,d\Omega^i,$$

and

$$(P_\ell)_k = \frac{1}{8\pi}\lim_{r\to\infty}\int_{S_r} (h_{ik} - \delta_{ik}h_{jj})\,d\Omega^i.$$

Substituting these into the integrated form of (4.18) gives exactly (4.22). □

The same boundary limit holds for any spinor which approaches ψ_∞ fast enough at infinity.

Proposition 4.23. *Let ψ be a smooth spinor on the end M_ℓ, and suppose that*

$$\psi = \psi_\infty + \eta$$

for some constant spinor ψ_∞, where

$$\eta = o(1), \qquad V\eta = o(r^{-2}) \qquad (r \to \infty).$$

Then

$$\lim_{r\to\infty} \int_{S_r} \beta(\psi) = \lim_{r\to\infty} \int_{S_r} \beta(\psi_\infty). \tag{4.24}$$

In particular, (4.22) remains valid with ψ in place of ψ_∞.

Proof. Since β is bilinear in ψ and $V\psi$, we have

$$\beta(\psi) - \beta(\psi_\infty) = -\frac{1}{2}\Big(\langle \eta, [e^i, e^j]\cdot V_j\psi_\infty\rangle + \langle \psi_\infty, [e^i, e^j]\cdot V_j\eta\rangle + \langle \eta, [e^i, e^j]\cdot V_j\eta\rangle\Big) e_i \lrcorner \mu.$$

Therefore

$$|\beta(\psi) - \beta(\psi_\infty)| \le C\Big(|\eta|\,|V\psi_\infty| + |\psi_\infty|\,|V\eta| + |\eta|\,|V\eta|\Big)\, dS.$$

Now $V\psi_\infty = O(r^{-2})$ by (4.13), while $\eta = o(1)$ and $V\eta = o(r^{-2})$ by hypothesis. Thus the integrand is $o(r^{-2})\, dS$, and since $\mathrm{Area}(S_r) = O(r^2)$, the integral tends to zero as $r \to \infty$. □

We can finally isolate the factor $E_\ell - |P_\ell|$.

Proposition 4.25. *Define the endomorphism*

$$\mathcal{P}_\ell := (P_\ell)_k\, dx^0 \cdot dx^k$$

on the asymptotic spinor space over the end M_ℓ. Then:

(i) $\mathcal{P}_\ell$ is Hermitian with respect to $\langle\,\cdot\,,\,\cdot\,\rangle$;

(ii)

$$\mathcal{P}_\ell^2 = |P_\ell|^2\,\mathrm{Id};$$

(iii) the eigenvalues of $\mathcal{P}_\ell$ are $\pm|P_\ell|$;

(iv) if ψ_∞ is an eigenvector with eigenvalue $-|P_\ell|$, then

$$E_\ell\,\langle \psi_\infty, \psi_\infty\rangle + \big\langle \psi_\infty, \mathcal{P}_\ell\psi_\infty\big\rangle = (E_\ell - |P_\ell|)\,\langle \psi_\infty, \psi_\infty\rangle. \tag{4.26}$$

Proof. By Section 2, each operator $dx^0 \cdot dx^k$ is Hermitian with respect to $\langle\,\cdot\,,\,\cdot\,\rangle$, so $\mathcal{P}_\ell$ is Hermitian.

Next,

$$\mathcal{P}_\ell^2 = (P_\ell)_k (P_\ell)_m\, dx^0 \cdot dx^k \cdot dx^0 \cdot dx^m.$$

Since dx^0 anticommutes with dx^k and $(dx^0)^2 = \mathrm{Id}$,

$$dx^0 \cdot dx^k \cdot dx^0 = -dx^k,$$

hence

$$\mathcal{P}_\ell^2 = -(P_\ell)_k (P_\ell)_m\, dx^k \cdot dx^m.$$

Splitting into diagonal and off-diagonal parts and using

$$dx^k \cdot dx^k = -\mathrm{Id}, \qquad dx^k \cdot dx^m + dx^m \cdot dx^k = 0 \quad (k \ne m),$$

we obtain

$$\mathcal{P}_\ell^2 = \sum_{k=1}^{3} (P_\ell)_k^2\,\mathrm{Id} = |P_\ell|^2\,\mathrm{Id}.$$

This proves (ii). Since $\mathcal{P}_\ell$ is Hermitian and satisfies the polynomial equation

$$\lambda^2 - |P_\ell|^2 = 0,$$

its eigenvalues are $\pm|P_\ell|$. Finally, if $\mathcal{P}_\ell\psi_\infty = -|P_\ell|\psi_\infty$, then

$$\left\langle \psi_\infty, \mathcal{P}_\ell\psi_\infty \right\rangle = -|P_\ell| \left\langle \psi_\infty, \psi_\infty \right\rangle,$$

which is exactly (4.26). □

Combining Proposition 4.23 with Proposition 4.25, we find that if a spinor ψ approaches a constant eigenspinor ψ_∞ on the end M_ℓ, and if ψ_∞ is chosen so that

$$\mathcal{P}_\ell\psi_\infty = -|P_\ell|\psi_\infty,$$

then

$$\lim_{r\to\infty} \int_{S_r} \beta(\psi) = 4\pi \left(E_\ell - |P_\ell|\right) \left\langle \psi_\infty, \psi_\infty \right\rangle. \tag{4.27}$$

This is the boundary expression that will be inserted into the integral identity in the proof of the positive energy theorem.

5. Weighted Sobolev spaces and the analytic theorem

In Sections 3 and 4, the positive energy theorem was reduced to the existence of solutions of the hypersurface Dirac equation

$$\mathscr{D}\psi = 0$$

with prescribed asymptotically constant behavior on the ends. The point of the present section is to establish the analytic theorem that provides these solutions.

The operator $\mathscr{D}$ is elliptic, but the manifold M is noncompact, so one must work in function spaces which simultaneously record one derivative and a definite rate of decay on each asymptotically flat end. Following Parker–Taubes, we use the weighted Sobolev spaces $\mathcal{H}_{s,\delta,p}$.

5.1. The weighted spaces $\mathcal{H}_{s,\delta,p}$.

Fix $R > 1$ large enough so that each asymptotic end M_ℓ contains the exterior of the Euclidean ball $B_R \subset \mathbb{R}^3$ under the chosen asymptotic chart. For each end M_ℓ and each $r > R$, write

$$M_{\ell,r} := M_\ell \cap \{x \in \mathbb{R}^3 : \ |x| > r\}.$$

Choose once and for all a smooth function

$$\sigma : M \longrightarrow [1, \infty)$$

such that

(i) $\sigma \geq 1$ everywhere on M,
(ii) $\sigma = r$ on each exterior region $M_{\ell,2R}$,
(iii) $\sigma \equiv 1$ on the compact set $M \setminus \bigcup_\ell M_{\ell,R}$.

Thus σ agrees with the Euclidean radial variable on each end for large r, while staying uniformly bounded away from zero on the compact core.

Definition 5.1. Let $p \geq 2$, let $s \in \{0, 1\}$, and let δ satisfy

$$\frac{1}{2} - \frac{3}{p} \leq \delta \leq 2 - \frac{3}{p}. \tag{5.2}$$

The weighted Sobolev space $\mathcal{H}_{s,\delta,p}(M;S)$ is the completion of $C_0^\infty(M;S)$ in the norm

$$\|\psi\|_{s,\delta,p} = \begin{cases} \|\sigma^\delta \psi\|_{L^p}, & s=0, \\ \|\sigma^{1+\delta}\nabla\psi\|_{L^p} + \|\sigma^\delta\psi\|_{L^p}, & s=1. \end{cases} \tag{5.3}$$

Here ∇ is the intrinsic Riemannian spin connection from Section 2. The use of ∇ rather than V in (5.3) is natural because weighted Sobolev spaces should be defined in terms of the intrinsic geometry of (M,g); the extrinsic contribution will enter later through the fact that $V-\nabla$ is a lower-order term.

A few remarks are immediate.

First, $\mathcal{H}_{0,\delta,p}$ measures the L^p-decay of ψ weighted by σ^δ. Thus larger δ corresponds to stronger decay. Second, $\mathcal{H}_{1,\delta,p}$ requires one weighted derivative as well. Third, the lower bound in (5.2) is exactly what makes the weighted L^p-norm strong enough to imply L^2-control, while the upper bound is the one that later matches the Euclidean end estimates for first-order elliptic operators.

For notational convenience, we write

$$\mathcal{H} := \mathcal{H}_{1,-1,2}, \qquad \|\psi\|_{\mathcal{H}}^2 := \|V\psi\|_{L^2}^2 + \|\sigma^{-1}\psi\|_{L^2}^2.$$

In the Hilbert case $(p,\delta)=(2,-1)$, this norm will play a central role.

5.2. **Mapping properties of V and $\mathscr{D}$.** The first step is to show that the ambient spin connection V and the hypersurface Dirac operator $\mathscr{D}$ act continuously between the weighted spaces suggested by their order.

Recall from Proposition 3.9 that in a local orthonormal frame,

$$V_i\psi = \nabla_i\psi + \frac{1}{2}h_{ij}e^0\cdot e^j\cdot\psi.$$

Thus the difference $V-\nabla$ is a zeroth-order term with coefficient h_{ij}. Since $h_{ij} = O(r^{-2})$ on the ends and $\sigma = r$ there for large r, we have

$$|h| \le C\,\sigma^{-2} \tag{5.4}$$

for some constant $C>0$ on all of M.

Proposition 5.5. *Let $p\ge 2$, and assume either*

$$0<\delta<2-\frac{3}{p}, \qquad \text{or} \qquad (p,\delta)=(2,-1).$$

Then

$$V:\mathcal{H}_{1,\delta,p}\longrightarrow\mathcal{H}_{0,\delta+1,p}$$

and

$$\mathscr{D}:\mathcal{H}_{1,\delta,p}\longrightarrow\mathcal{H}_{0,\delta+1,p}$$

are bounded linear maps.

Proof. We begin with V. By Proposition 3.9,

$$|V\psi| \le |\nabla\psi| + C|h|\,|\psi|.$$

Multiplying by $\sigma^{1+\delta}$ and using (5.4),

$$\sigma^{1+\delta}|V\psi| \le \sigma^{1+\delta}|\nabla\psi| + C\,\sigma^{1+\delta}\sigma^{-2}|\psi| = \sigma^{1+\delta}|\nabla\psi| + C\,\sigma^{\delta-1}|\psi|.$$

Since $\sigma\ge 1$, we have $\sigma^{\delta-1}\le\sigma^\delta$. Hence

$$\|\sigma^{1+\delta}V\psi\|_{L^p} \le C\big(\|\sigma^{1+\delta}\nabla\psi\|_{L^p} + \|\sigma^\delta\psi\|_{L^p}\big).$$

This is exactly the boundedness of V.

For $\mathscr{D}$, write

$$\mathscr{D}\psi = \sum_{i=1}^{3} e^i \cdot V_i\psi.$$

Because Clifford multiplication by an orthonormal covector has pointwise norm 1,

$$|\mathscr{D}\psi| \le \sum_{i=1}^{3} |V_i\psi| \le \sqrt{3}\,|V\psi|.$$

Therefore

$$\|\sigma^{1+\delta}\mathscr{D}\psi\|_{L^p} \le \sqrt{3}\,\|\sigma^{1+\delta}V\psi\|_{L^p},$$

and the desired boundedness follows from the estimate already proved for V. □

The point of Proposition 5.5 is that the hypersurface Dirac operator behaves like a first-order elliptic operator with asymptotically decaying zeroth-order coefficients. The decay $h = O(r^{-2})$ is precisely what prevents the extra term in V from spoiling the weighted estimate.

5.3. **The basic L^2 identity and the vanishing theorem.** We next turn the integrated Weitzenböck formula from Section 4 into an L^2 identity for compactly supported spinors. This is the analytic form of the geometric positivity established earlier.

Lemma 5.6. *For every $\psi \in C_0^\infty(M;S)$,*

$$\|\mathscr{D}\psi\|_{L^2}^2 = \|V\psi\|_{L^2}^2 + \int_M \langle \psi, \mathcal{R}\psi\rangle\,\mu. \tag{5.7}$$

In particular, under the dominant energy condition,

$$\|\mathscr{D}\psi\|_{L^2} \ge \|V\psi\|_{L^2}. \tag{5.8}$$

Proof. Since ψ is compactly supported, the boundary term in the integrated Weitzenböck formula (4.5) vanishes. Thus

$$\int_M \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi\rangle - |\mathscr{D}\psi|^2\Big)\,\mu = 0,$$

which is (5.7). If the dominant energy condition holds, then $\langle \psi, \mathcal{R}\psi\rangle \ge 0$ by Proposition 3.39, and (5.8) follows. □

To use (5.7) beyond compactly supported spinors, we need a simple embedding into the Hilbert space $\mathcal{H}$.

Lemma 5.9. *Let $p \ge 2$ and δ satisfy*

$$\delta > \frac{1}{2} - \frac{3}{p}.$$

Then there is a continuous embedding

$$\mathcal{H}_{1,\delta,p} \hookrightarrow \mathcal{H}.$$

Moreover, for $\psi \in \mathcal{H}_{1,\delta,p}$,

$$\|\sigma^{-1}\psi\|_{L^2} + \|V\psi\|_{L^2} \le C\,\|\psi\|_{1,\delta,p}. \tag{5.10}$$

Proof. By Hölder's inequality,

$$\|\sigma^{-1}\psi\|_{L^2} = \|\sigma^{-(1+\delta)}\,\sigma^{\delta}\psi\|_{L^2} \le \|\sigma^{-(1+\delta)}\|_{L^{\frac{2p}{p-2}}}\,\|\sigma^{\delta}\psi\|_{L^p}.$$

The exponent condition

$$(1+\delta)\frac{2p}{p-2} > 3$$

is equivalent to $\delta > \frac{1}{2} - \frac{3}{p}$, so the first factor is finite. Hence

$$\|\sigma^{-1}\psi\|_{L^2} \le C\|\sigma^{\delta}\psi\|_{L^p}.$$

Similarly,

$$\|V\psi\|_{L^2} = \|\sigma^{-(1+\delta)}\,\sigma^{1+\delta}V\psi\|_{L^2} \le \|\sigma^{-(1+\delta)}\|_{L^{\frac{2p}{p-2}}}\,\|\sigma^{1+\delta}V\psi\|_{L^p}.$$

Using Proposition 5.5,

$$\|\sigma^{1+\delta}V\psi\|_{L^p} \le C\|\psi\|_{1,\delta,p}.$$

Combining these inequalities yields (5.10). □

We can now prove the vanishing theorem, which is the analytic uniqueness statement underlying the Green's function construction. The point is exactly the one emphasized by Parker and Taubes: from $\mathscr{D}\psi = 0$ and the Weitzenböck formula one gets $V\psi = 0$, but since V is not compatible with the positive definite inner product, vanishing does *not* follow automatically. One must use the decay of the second fundamental form.

Proposition 5.11 (Vanishing theorem)**.** *Let $p \ge 2$, and assume either*

$$0 < \delta < 2 - \frac{3}{p}, \qquad \text{or} \qquad (p,\delta) = (2,-1).$$

If $\psi \in \mathcal{H}_{1,\delta,p}$ satisfies

$$\mathscr{D}\psi = 0,$$

then $\psi \equiv 0$.

Proof. By elliptic regularity for first-order elliptic systems, ψ is smooth. Let $\{\psi_n\} \subset C_0^\infty(M;S)$ be a sequence converging to ψ in $\mathcal{H}_{1,\delta,p}$. By Lemma 5.9, $\psi_n \to \psi$ in $\mathcal{H}$, and by Proposition 5.5, $\mathscr{D}\psi_n \to \mathscr{D}\psi = 0$ in L^2. Passing to the limit in (5.7) gives

$$0 = \|V\psi\|_{L^2}^2 + \int_M \langle \psi, \mathcal{R}\psi\rangle\,\mu.$$

Since both terms are nonnegative under the dominant energy condition, we conclude that

$$V\psi = 0.$$

We now use the decay information carried by $\psi \in \mathcal{H}_{1,\delta,p}$. On each end M_ℓ,

$$\int_{M_{\ell,2R}} \sigma^{p\delta}|\psi|^p\,\mu < \infty.$$

Since $\sigma = r$ on the end, Fubini's theorem implies that along almost every ray $\omega \in S^2$,

$$\int_{2R}^{\infty} r^{p\delta+2}|\psi(r,\omega)|^p\,dr < \infty,$$

and therefore $|\psi(r,\omega)|$ takes arbitrarily small values along that ray as $r \to \infty$.

It remains to show that a V-parallel spinor that becomes arbitrarily small along one path to infinity must vanish identically. Set

$$u := |\psi| = \langle \psi, \psi \rangle^{1/2}.$$

On the open set where $u \neq 0$, differentiate u^2:

$$d(u^2)(e_i) = e_i \langle \psi, \psi \rangle.$$

By Lemma 3.14 and $V_i \psi = 0$,

$$e_i \langle \psi, \psi \rangle = \langle h_{ij} e^j \cdot e^0 \cdot \psi, \psi \rangle.$$

Hence

$$|d(u^2)| \leq |h|\, u^2.$$

Since $d(u^2) = 2u\, du$, we obtain on the set $\{u \neq 0\}$,

$$|d(\log u)| \leq \frac{1}{2}|h|.$$

Using $h = O(r^{-2})$, there is a constant C such that on each end

$$|d(\log u)| \leq C r^{-2}.$$

Let $\gamma : [0, \infty) \to M_\ell$ be a path to infinity along which u takes arbitrarily small values. Then for $0 < s < t$,

$$\big|\log u(\gamma(t)) - \log u(\gamma(s))\big| \leq C \int_s^t r(\gamma(\tau))^{-2}\, |\gamma'(\tau)|\, d\tau.$$

The integral on the right converges as $t \to \infty$. Therefore if u approaches arbitrarily small values along γ, it must vanish identically along the path. Since $V\psi = 0$, parallel transport then shows that ψ vanishes everywhere on the connected manifold M. Thus $\psi \equiv 0$. □

Proposition 5.11 is the uniqueness statement needed both in the Hilbert-space construction and in the weighted L^p theory.

5.4. **The coercive estimate in the Hilbert case.** We now single out the Hilbert case $(p, \delta) = (2, -1)$. This is where one first obtains a genuine inverse for $\mathscr{D}$, via a coercive estimate.

Recall that

$$\mathcal{H} = \mathcal{H}_{1,-1,2}, \qquad \|\psi\|_{\mathcal{H}}^2 = \|V\psi\|_{L^2}^2 + \|\sigma^{-1}\psi\|_{L^2}^2.$$

The next lemma controls the weighted L^2 part on each exterior region.

Lemma 5.12. *There exists $R_0 > 0$ such that for every $R \geq R_0$, every end M_ℓ, and every $\psi \in \mathcal{H}$,*

$$\|\sigma^{-1}\psi\|_{L^2(M_{\ell,2R})} \leq 5\, \|d|\psi|\|_{L^2(M_{\ell,2R})}, \tag{5.13}$$

and

$$\|\sigma^{-1}\psi\|_{L^2(M_{\ell,2R})} + \|\nabla\psi\|_{L^2(M_{\ell,2R})} \leq C\, \|V\psi\|_{L^2(M_{\ell,2R})}. \tag{5.14}$$

Proof. Fix one end and identify it with a subset of $\mathbb{R}^3$. For compactly supported smooth spinors, the Euclidean Hardy inequality gives

$$\int_{|x|>2R} r^{-2}|\psi|^2\, d\mu_3 \le 4\int_{|x|>2R} |d|\psi||^2\, d\mu_3,$$

where $d\mu_3$ is Euclidean measure. Since $g_{ij} = \delta_{ij} + O(r^{-1})$, the Riemannian volume form is comparable to $d\mu_3$ on the end, and (5.13) follows, after increasing the constant slightly.

Next, Kato's inequality gives

$$|d|\psi|| \le |\nabla\psi|.$$

Also,

$$V_i\psi = \nabla_i\psi + \frac{1}{2}h_{ij}e^0\cdot e^j\cdot\psi,$$

so

$$|\nabla\psi| \le |V\psi| + C|h|\,|\psi|.$$

Using $|h| \le Cr^{-2}$ and (5.13),

$$\|\nabla\psi\|_{L^2(M_{\ell,2R})} \le \|V\psi\|_{L^2(M_{\ell,2R})} + C\|r^{-2}\psi\|_{L^2(M_{\ell,2R})} \le \|V\psi\|_{L^2(M_{\ell,2R})} + CR^{-1}\|r^{-1}\psi\|_{L^2(M_{\ell,2R})}.$$

For R large, the last term can be absorbed into the left-hand side, yielding (5.14). □

We now globalize the end estimate by cutting a spinor into interior and exterior pieces.

Proposition 5.15 (Coercive estimate)**.** *There exists a constant $c > 0$ such that for all $\psi \in \mathcal{H}$,*

$$\|\psi\|_{\mathcal{H}}^2 \le c\,\|\mathscr{D}\psi\|_{L^2}^2. \tag{5.16}$$

Proof. Choose R large enough that Lemma 5.12 holds. Let $\beta = \beta_R$ be a smooth cutoff function such that

(i) $0 \le \beta \le 1$,
(ii) $\beta \equiv 1$ on each end $M_{\ell,3R}$,
(iii) $\beta \equiv 0$ on the compact core

$$M_0 := M \setminus \bigcup_{\ell} M_{\ell,2R},$$

(iv) $|d\beta| \le C\sigma^{-1}$.

Write

$$\psi = \psi_{\text{in}} + \psi_{\text{ex}}, \qquad \psi_{\text{in}} := (1-\beta)\psi, \qquad \psi_{\text{ex}} := \beta\psi.$$

We first estimate the interior part. Since ψ_{in} is supported in the fixed compact set M_0, standard elliptic estimates on compact manifolds give

$$\|\psi_{\text{in}}\|_{H^1(M_0)} \le C\,\|V\psi_{\text{in}}\|_{L^2(M_0)}.$$

Because $\sigma \equiv 1$ on M_0, this implies

$$\|\sigma^{-1}\psi_{\text{in}}\|_{L^2} + \|V\psi_{\text{in}}\|_{L^2} \le C\,\|V\psi_{\text{in}}\|_{L^2} \le C\,\|V\psi\|_{L^2}. \tag{5.17}$$

For the exterior part, Lemma 5.12 yields

$$\|\sigma^{-1}\psi_{\text{ex}}\|_{L^2} + \|\nabla\psi_{\text{ex}}\|_{L^2} \le C\,\|V\psi_{\text{ex}}\|_{L^2}.$$

Now

$$V\psi_{\mathrm{ex}} = d\beta \otimes \psi + \beta\, V\psi.$$

Since $|d\beta| \le C\sigma^{-1}$,

$$\|V\psi_{\mathrm{ex}}\|_{L^2} \le C\big(\|\sigma^{-1}\psi\|_{L^2} + \|V\psi\|_{L^2}\big).$$

Combining with the previous inequality and then with (5.17), we obtain

$$\|\sigma^{-1}\psi\|_{L^2} \le C\|V\psi\|_{L^2}.$$

Hence

$$\|\psi\|_{\mathcal{H}}^2 = \|V\psi\|_{L^2}^2 + \|\sigma^{-1}\psi\|_{L^2}^2 \le C\|V\psi\|_{L^2}^2.$$

Finally, Lemma 5.6 gives

$$\|V\psi\|_{L^2}^2 \le \|\mathscr{D}\psi\|_{L^2}^2,$$

so (5.16) follows. □

The significance of Proposition 5.15 is that it converts the geometric positivity from Section 4 into an honest Hilbert-space estimate. This is what makes Lax–Milgram available.

5.5. **The weighted a priori estimate.** We first solve $\mathscr{D}\psi = \eta$ for compactly supported η, and then derive the weighted L^p estimate needed for the full isomorphism theorem.

Proposition 5.18. *For each $\eta \in C_0^\infty(M;S)$, there exists a unique $\psi \in \mathcal{H}$ such that*

$$\mathscr{D}\psi = \eta.$$

Moreover,

$$\|\psi\|_{\mathcal{H}} \le C\|\eta\|_{L^2}. \tag{5.19}$$

Proof. Consider the bilinear form

$$B(u,v) := \int_M \langle \mathscr{D}u, \mathscr{D}v\rangle\, \mu$$

on the Hilbert space $\mathcal{H}$. By Proposition 5.15,

$$B(u,u) = \|\mathscr{D}u\|_{L^2}^2 \ge c^{-1}\|u\|_{\mathcal{H}}^2,$$

so B is coercive. Also B is continuous on $\mathcal{H}$.

Since η is compactly supported, the linear functional

$$L_\eta(v) := \int_M \langle \eta, v\rangle\, \mu$$

is continuous on $\mathcal{H}$. By Lax–Milgram, there exists a unique $u \in \mathcal{H}$ such that

$$B(u,v) = L_\eta(v) \qquad \text{for all } v \in \mathcal{H}.$$

Equivalently,

$$\int_M \langle \mathscr{D}u, \mathscr{D}v\rangle\, \mu = \int_M \langle \eta, v\rangle\, \mu.$$

Set

$$\psi := \mathscr{D}u.$$

Then $\psi \in \mathcal{H}$, and in the weak sense

$$\mathscr{D}\psi = \mathscr{D}^2 u = \eta.$$

By elliptic regularity, ψ is smooth and satisfies the equation strongly. Uniqueness follows from Proposition 5.11. The estimate (5.19) comes from coercivity. □

We now turn to the weighted L^p estimate. This is the part of the proof where the noncompact geometry of the ends enters in an essential way. On the compact interior one has the usual elliptic estimates. On the ends one compares $\mathscr{D}$ to a first-order operator on $\mathbb{R}^3$ obtained by cutting off the metric and the second fundamental form. This is exactly where the Euclidean model theorem from the appendix is used.

Proposition 5.20 (Weighted a priori estimate)**.** *Let $p \geq 2$, and assume either*

$$0 < \delta < 2 - \frac{3}{p}, \qquad \text{or} \qquad (p, \delta) = (2, -1).$$

If $\eta \in C_0^\infty(M; S)$ and $\psi \in \mathcal{H}$ satisfies

$$\mathscr{D}\psi = \eta,$$

then

$$\|\psi\|_{1,\delta,p} \leq C(\delta, p)\, \|\eta\|_{0,\delta+1,p}. \tag{5.21}$$

Proof. We split ψ as above into

$$\psi = \psi_{\text{in}} + \psi_{\text{ex}}$$

using a cutoff β supported on the ends.

Step 1: the interior estimate. Since ψ_{in} is supported in a fixed compact set, standard interior elliptic estimates for first-order operators imply

$$\|\psi_{\text{in}}\|_{1,\delta,p} \leq C\big(\|\mathscr{D}\psi_{\text{in}}\|_{0,\delta+1,p} + \|\psi_{\text{in}}\|_{L^p}\big). \tag{5.22}$$

On the compact set where ψ_{in} is supported, the weight σ is bounded above and below, so the weighted and unweighted norms are equivalent. Moreover,

$$\mathscr{D}\psi_{\text{in}} = (1 - \beta)\eta - (d\beta) \cdot \psi,$$

hence

$$\|\mathscr{D}\psi_{\text{in}}\|_{0,\delta+1,p} \leq C\big(\|\eta\|_{0,\delta+1,p} + \|\sigma^{-1}\psi\|_{L^p}\big).$$

Since $\psi \in \mathcal{H}$ and $p \geq 2$, the compact support of $d\beta$ gives

$$\|\sigma^{-1}\psi\|_{L^p(\operatorname{supp} d\beta)} \leq C\|\psi\|_{\mathcal{H}}.$$

Thus

$$\|\psi_{\text{in}}\|_{1,\delta,p} \leq C\big(\|\eta\|_{0,\delta+1,p} + \|\psi\|_{\mathcal{H}}\big). \tag{5.23}$$

Step 2: the exterior estimate. Fix one end M_ℓ. Using the asymptotic chart and the trivialization of S on the end, extend ψ_{ex} and η to compactly supported spinors on $\mathbb{R}^3$. Also extend the metric g and the second fundamental form h to $\mathbb{R}^3$ by cutting them off outside a sufficiently large ball. This produces a first-order elliptic operator $\mathscr{D}_R$ on $\mathbb{R}^3$ which agrees with $\mathscr{D}$ on the region where ψ_{ex} is supported.

By the Euclidean end theorem proved in Appendix B, one may choose the cutoff radius R so large that

$$\mathscr{D}_R : \mathcal{H}_{1,\delta,p}(\mathbb{R}^3; S) \longrightarrow \mathcal{H}_{0,\delta+1,p}(\mathbb{R}^3; S)$$

is an isomorphism with bounded inverse. Therefore

$$\|\psi_{\text{ex}}\|_{1,\delta,p} \leq C(\delta, p)\, \|\mathscr{D}_R\psi_{\text{ex}}\|_{0,\delta+1,p}.$$

Since $\mathscr{D}_R$ agrees with $\mathscr{D}$ on the support of ψ_{ex},

$$\mathscr{D}_R\psi_{\mathrm{ex}} = \beta\eta + (d\beta)\cdot\psi.$$

Hence

$$\|\psi_{\mathrm{ex}}\|_{1,\delta,p} \le C(\delta,p)\big(\|\eta\|_{0,\delta+1,p} + \|(d\beta)\cdot\psi\|_{0,\delta+1,p}\big).$$

As before, $|d\beta| \le C\sigma^{-1}$ and $d\beta$ is supported in a fixed compact annulus, so

$$\|(d\beta)\cdot\psi\|_{0,\delta+1,p} \le C\|\psi\|_{\mathcal{H}}.$$

Thus

$$\|\psi_{\mathrm{ex}}\|_{1,\delta,p} \le C(\delta,p)\big(\|\eta\|_{0,\delta+1,p} + \|\psi\|_{\mathcal{H}}\big). \tag{5.24}$$

Step 3: combine the two pieces. Adding (5.23) and (5.24),

$$\|\psi\|_{1,\delta,p} \le C(\delta,p)\big(\|\eta\|_{0,\delta+1,p} + \|\psi\|_{\mathcal{H}}\big).$$

Finally, Proposition 5.18 gives

$$\|\psi\|_{\mathcal{H}} \le C\|\eta\|_{L^2}.$$

Since η is compactly supported and $p \ge 2$, the compactness of its support implies

$$\|\eta\|_{L^2} \le C\|\eta\|_{0,\delta+1,p}.$$

Substituting this into the previous inequality yields (5.21). □

5.6. **The isomorphism theorem for $\mathscr{D}$.** We now complete the proof of the analytic theorem.

Proposition 5.25 (Surjectivity)**.** *Let $p \ge 2$, and assume either*

$$0 < \delta < 2 - \frac{3}{p}, \qquad \text{or} \qquad (p,\delta) = (2,-1).$$

Then

$$\mathscr{D} : \mathcal{H}_{1,\delta,p} \longrightarrow \mathcal{H}_{0,\delta+1,p}$$

is surjective. Moreover, if $\eta \in \mathcal{H}_{0,\delta+1,p}$ and $\mathscr{D}\psi = \eta$, then

$$\|\psi\|_{1,\delta,p} \le C(\delta,p)\,\|\eta\|_{0,\delta+1,p}. \tag{5.26}$$

Proof. Let $\eta \in \mathcal{H}_{0,\delta+1,p}$. Choose a sequence

$$\eta_m \in C_0^\infty(M;S)$$

converging strongly to η in $\mathcal{H}_{0,\delta+1,p}$. By Proposition 5.18, for each m there exists a unique $\psi_m \in \mathcal{H}$ solving

$$\mathscr{D}\psi_m = \eta_m.$$

By Proposition 5.20,

$$\|\psi_m - \psi_n\|_{1,\delta,p} \le C(\delta,p)\,\|\eta_m - \eta_n\|_{0,\delta+1,p}.$$

Hence $\{\psi_m\}$ is Cauchy in $\mathcal{H}_{1,\delta,p}$. Let

$$\psi := \lim_{m\to\infty} \psi_m \qquad \text{in } \mathcal{H}_{1,\delta,p}.$$

Since $\mathscr{D}$ is continuous by Proposition 5.5,

$$\mathscr{D}\psi = \lim_{m\to\infty} \mathscr{D}\psi_m = \lim_{m\to\infty} \eta_m = \eta.$$

Thus $\mathscr{D}$ is surjective.

The estimate (5.26) follows by passing to the limit in (5.21). Uniqueness of the solution in $\mathcal{H}_{1,\delta,p}$ follows from Proposition 5.11. □

We can now collect the results.

Theorem 5.27 (Analytic theorem for the hypersurface Dirac operator)**.** *Let $p \geq 2$. Then:*

(i) if $0 < \delta < 2 - \frac{3}{p}$, the operator

$$\mathscr{D} : \mathcal{H}_{1,\delta,p} \longrightarrow \mathcal{H}_{0,\delta+1,p}$$

is an isomorphism with bounded inverse;

(ii) in the Hilbert case $(p, \delta) = (2, -1)$, the operator

$$\mathscr{D} : \mathcal{H}_{1,-1,2} \longrightarrow \mathcal{H}_{0,0,2}$$

is also an isomorphism with bounded inverse;

(iii) if $\eta \in \mathcal{H}_{0,\delta+1,p} \cap C^\infty(M; S)$, then the unique solution $\psi \in \mathcal{H}_{1,\delta,p}$ of

$$\mathscr{D}\psi = \eta$$

is smooth.

Proof. Injectivity is Proposition 5.11. Surjectivity and the a priori estimate are Proposition 5.25. Boundedness of the inverse is exactly the estimate (5.26). Finally, smoothness follows from elliptic regularity. □

Theorem 5.27 is the analytic input needed in the geometric proof. To construct a Witten spinor with prescribed asymptotic constants, one starts with a cutoff asymptotic spinor ψ_0, observes that $\mathscr{D}\psi_0$ lies in the appropriate weighted space, and then solves

$$\mathscr{D}\psi_1 = -\mathscr{D}\psi_0$$

by Theorem 5.27. The actual construction is short, and we carry it out in the next section.

6. Construction of Witten spinors

We now combine the geometric work of Sections 3 and 4 with the analytic theorem from Section 5. The outcome is the existence of harmonic spinors on M with prescribed constant values on the asymptotic ends. These are the Witten spinors that will be inserted into the boundary formula in the next section.

6.1. Cutoff asymptotic spinors.

Fix, for each end M_ℓ, a constant spinor

$$\psi_{\infty,\ell}$$

in the distinguished asymptotic trivialization of S over M_ℓ. We shall construct a global smooth spinor ψ on M satisfying

$$\mathscr{D}\psi = 0$$

and

$$\psi \sim \psi_{\infty,\ell} \qquad \text{on } M_\ell.$$

Choose $R > 1$ large enough that every end contains the exterior of the Euclidean ball B_R, and let

$$M_{\ell,r} := M_\ell \cap \{x \in \mathbb{R}^3 : \ |x| > r\}.$$

For each ℓ, choose a smooth cutoff function

$$f_R^{(\ell)} : M \longrightarrow [0, 1]$$

such that

(i) $f_R^{(\ell)} \equiv 1$ on $M_{\ell,3R}$,
(ii) $f_R^{(\ell)} \equiv 0$ on $M \setminus M_{\ell,2R}$,
(iii) $|df_R^{(\ell)}| \le C\,\sigma^{-1}$.

Thus $f_R^{(\ell)}$ vanishes off the ℓ-th end, becomes identically 1 far out on that end, and changes only in the fixed annular region

$$M_{\ell,2R} \setminus M_{\ell,3R}.$$

Definition 6.1. Define the cutoff asymptotic spinor

$$\psi_0 := \sum_{\ell=1}^{k} f_R^{(\ell)}\,\psi_{\infty,\ell}. \tag{6.2}$$

By construction, ψ_0 is a smooth global spinor on M. On the ℓ-th end, for $r > 3R$,

$$\psi_0 = \psi_{\infty,\ell},$$

while on the compact core $M \setminus \bigcup_\ell M_{\ell,2R}$ we have

$$\psi_0 = 0.$$

Thus ψ_0 has exactly the desired asymptotic constants, but it is not yet harmonic.

The next step is to compute $\mathscr{D}\psi_0$.

Lemma 6.3. *On each end M_ℓ,*

$$\mathscr{D}\psi_0 = O(r^{-2}). \tag{6.4}$$

Consequently, for every $p \ge 2$ and every δ satisfying

$$0 < \delta < 1 - \frac{3}{p}, \tag{6.5}$$

one has

$$\mathscr{D}\psi_0 \in \mathcal{H}_{0,\delta+1,p}(M;S). \tag{6.6}$$

Proof. Fix one end M_ℓ. There are two regions to consider.

Region 1: $r > 3R$. Here $f_R^{(\ell)} \equiv 1$, so $\psi_0 = \psi_{\infty,\ell}$. Since $\psi_{\infty,\ell}$ is constant in the asymptotic trivialization, the argument of Section 4 shows that

$$V\psi_{\infty,\ell} = O(r^{-2}).$$

Therefore

$$\mathscr{D}\psi_0 = \mathscr{D}\psi_{\infty,\ell} = \sum_{i=1}^{3} e^i \cdot V_i\psi_{\infty,\ell} = O(r^{-2}).$$

Region 2: $2R \le r \le 3R$. In this annulus,

$$\psi_0 = f_R^{(\ell)}\psi_{\infty,\ell},$$

so

$$\mathscr{D}\psi_0 = e^i \cdot V_i(f_R^{(\ell)}\psi_{\infty,\ell}) = e^i \cdot \big((e_i f_R^{(\ell)})\psi_{\infty,\ell} + f_R^{(\ell)} V_i\psi_{\infty,\ell}\big).$$

The term involving $V_i\psi_{\infty,\ell}$ is $O(r^{-2})$ as above. For the cutoff term, $|df_R^{(\ell)}| \le C\sigma^{-1}$, and on this fixed annulus $\sigma \sim R$, so $|df_R^{(\ell)}| = O(R^{-1})$. Since the annulus is contained in a compact region once R is fixed, this term is smooth and compactly supported. In particular it lies in every weighted L^p space. Thus (6.4) holds on the noncompact

part of the end, and globally $\mathscr{D}\psi_0$ differs from an $O(r^{-2})$ field only by a compactly supported smooth term.

We now verify (6.6). On each end, for r large,

$$|\mathscr{D}\psi_0| \le Cr^{-2}, \qquad \sigma = r,$$

so

$$|\sigma^{\delta+1}\mathscr{D}\psi_0|^p \le Cr^{p(\delta+1)}r^{-2p}.$$

Including the Euclidean volume element $r^2\,dr\,d\omega$, the relevant radial integral is bounded by

$$\int_R^\infty r^{p(\delta+1)-2p+2}\,dr = \int_R^\infty r^{p\delta-p+2}\,dr.$$

This converges at infinity exactly when

$$p\delta - p + 2 < -1,$$

that is,

$$\delta < 1 - \frac{3}{p}.$$

Since the compactly supported part contributes no difficulty, we obtain (6.6). □

Lemma 6.3 is the precise point at which the asymptotically constant boundary condition becomes compatible with the weighted spaces from Section 5.

6.2. **Solving the correction equation.** Choose $p > 3$ and δ such that

$$0 < \delta < 1 - \frac{3}{p}. \tag{6.7}$$

Then Lemma 6.3 gives

$$\mathscr{D}\psi_0 \in \mathcal{H}_{0,\delta+1,p}(M;S).$$

By Theorem 5.27, the operator

$$\mathscr{D} : \mathcal{H}_{1,\delta,p} \longrightarrow \mathcal{H}_{0,\delta+1,p}$$

is an isomorphism. Therefore there exists a unique spinor

$$\psi_1 \in \mathcal{H}_{1,\delta,p}(M;S) \tag{6.8}$$

such that

$$\mathscr{D}\psi_1 = -\mathscr{D}\psi_0. \tag{6.9}$$

Definition 6.10. Define

$$\psi := \psi_0 + \psi_1. \tag{6.11}$$

Proposition 6.12. *The spinor ψ defined by (6.11) is smooth and satisfies*

$$\mathscr{D}\psi = 0. \tag{6.13}$$

Moreover, on each end M_ℓ,

$$\psi - \psi_{\infty,\ell} = \psi_1 \qquad \text{for } r > 3R. \tag{6.14}$$

Proof. By (6.9),

$$\mathscr{D}\psi = \mathscr{D}\psi_0 + \mathscr{D}\psi_1 = \mathscr{D}\psi_0 - \mathscr{D}\psi_0 = 0.$$

Since $\mathscr{D}\psi_0$ is smooth and $\psi_1 \in \mathcal{H}_{1,\delta,p}$ solves a first-order elliptic equation, elliptic regularity implies that ψ_1 is smooth. Hence ψ is smooth as well.

Finally, on the region $M_{\ell,3R}$, the cutoff construction gives

$$\psi_0 = \psi_{\infty,\ell},$$

so

$$\psi - \psi_{\infty,\ell} = (\psi_0 + \psi_1) - \psi_{\infty,\ell} = \psi_1.$$

This is (6.14). □

The spinor ψ is the desired harmonic spinor with prescribed asymptotic constants, provided we can now show that ψ_1 decays to zero.

6.3. **Decay and uniqueness.** The weighted space $\mathcal{H}_{1,\delta,p}$ with $p > 3$ controls not only one derivative in L^p, but also pointwise decay through Sobolev embedding on the ends.

Proposition 6.15. *Let $p > 3$ and $0 < \delta < 1 - \frac{3}{p}$, and let ψ be the harmonic spinor constructed above. Then on each end M_ℓ,*

$$\psi(x) - \psi_{\infty,\ell} \longrightarrow 0 \qquad \text{as } r(x) \to \infty. \tag{6.16}$$

More precisely, if $\alpha = 1 - \frac{3}{p}$, then $\psi_1 \in \mathcal{H}_{1,\delta,p}$ has a Hölder continuous representative on each end, and

$$\psi - \psi_{\infty,\ell} = o(r^{-\delta}).$$

Proof. By Proposition 6.12, on the exterior region $M_{\ell,3R}$ we have

$$\psi - \psi_{\infty,\ell} = \psi_1.$$

Since $\psi_1 \in \mathcal{H}_{1,\delta,p}$ and $p > 3$, the weighted Sobolev embedding on asymptotically flat ends implies that ψ_1 has a Hölder continuous representative and tends to zero at infinity. More concretely, on each dyadic annulus

$$A_m := \{2^m \le r \le 2^{m+1}\},$$

the weighted bound

$$\|\sigma^{1+\delta}\nabla\psi_1\|_{L^p(A_m)} + \|\sigma^\delta\psi_1\|_{L^p(A_m)} \le C$$

rescales to a uniform $W^{1,p}$ bound for $r^\delta\psi_1$ on a fixed annulus. Since $p > 3$, Morrey's inequality gives a uniform Hölder bound for $r^\delta\psi_1$, and the weighted L^p-term forces its supremum on A_m to tend to zero as $m \to \infty$. Hence

$$r^\delta|\psi_1(r,\omega)| \longrightarrow 0,$$

which is exactly

$$\psi_1 = o(r^{-\delta}).$$

Therefore $\psi - \psi_{\infty,\ell} \to 0$ on each end. □

This proves the first two parts of Parker–Taubes' existence theorem: there exists a smooth harmonic spinor with the prescribed asymptotic constants, and the difference from those constants decays on each end.

We also record the uniqueness statement in the form needed later.

Proposition 6.17. *Let $p > 3$ and $0 < \delta < 1 - \frac{3}{p}$. Suppose ψ and $\widetilde{\psi}$ are smooth spinors on M such that*

$$\mathscr{D}\psi = \mathscr{D}\widetilde{\psi} = 0,$$

and such that on each end M_ℓ,

$$\psi - \psi_{\infty,\ell} \in \mathcal{H}_{1,\delta,p}, \qquad \widetilde{\psi} - \psi_{\infty,\ell} \in \mathcal{H}_{1,\delta,p}$$

for the same asymptotic constants $\psi_{\infty,\ell}$. Then

$$\psi = \widetilde{\psi}.$$

Proof. Set

$$u := \psi - \widetilde{\psi}.$$

Then

$$\mathscr{D}u = 0.$$

Moreover, on each end,

$$u = (\psi - \psi_{\infty,\ell}) - (\widetilde{\psi} - \psi_{\infty,\ell}),$$

so

$$u \in \mathcal{H}_{1,\delta,p}.$$

By Proposition 5.11, the only element of $\mathcal{H}_{1,\delta,p}$ annihilated by $\mathscr{D}$ is the zero spinor. Therefore

$$u \equiv 0,$$

hence $\psi = \widetilde{\psi}$. □

We summarize the conclusions of this section in a single theorem.

Theorem 6.18 (Witten spinors with prescribed asymptotic constants)**.** *Fix constant spinors*

$$\{\psi_{\infty,\ell}\}_{\ell=1}^{k}$$

on the asymptotic ends of M. Let $p > 3$ and choose

$$0 < \delta < 1 - \frac{3}{p}.$$

Then there exists a unique smooth spinor ψ on M such that

(i)

$$\mathscr{D}\psi = 0;$$

(ii) on each end M_ℓ,

$$\psi - \psi_{\infty,\ell} \in \mathcal{H}_{1,\delta,p};$$

(iii) in particular,

$$\psi(x) - \psi_{\infty,\ell} \longrightarrow 0 \qquad \text{as } r(x) \to \infty \text{ in } M_\ell.$$

Proof. Existence is the construction above. Smoothness is Proposition 6.12, the asymptotic decay is Proposition 6.15, and uniqueness is Proposition 6.17. □

Theorem 6.18 is exactly the analytic input needed for the positive energy theorem. In the next section, we insert these spinors into the integral identity from Section 4 and evaluate the boundary term using the prescribed asymptotic constants.

7. Proof of the positive energy theorem

We now assemble the ingredients from the previous sections. The argument is short. The existence of Witten spinors was proved in Section 6, the integral identity was established in Section 4, and the positivity of the zeroth-order term was proved in Section 3.

7.1. **Insert the Witten spinor into the integral identity.** Fix one asymptotic end M_ℓ. Choose constant spinors

$$\psi_{\infty,1}, \dots, \psi_{\infty,k}$$

on the ends of M by setting

$$\psi_{\infty,m} = 0 \qquad (m \neq \ell), \tag{7.1}$$

and leaving $\psi_{\infty,\ell}$ to be specified later.

By Theorem 6.18, there exists a unique smooth spinor

$$\psi \in \Gamma(S)$$

such that

$$\mathscr{D}\psi = 0, \tag{7.2}$$

and such that on each end M_m,

$$\psi - \psi_{\infty,m} \in \mathcal{H}_{1,\delta,p} \qquad \text{for some } p > 3,\ 0 < \delta < 1 - \frac{3}{p}. \tag{7.3}$$

In particular,

$$\psi(x) - \psi_{\infty,m} \longrightarrow 0 \qquad \text{as } r(x) \to \infty \text{ in } M_m.$$

Let

$$M_R := M \setminus \bigcup_{m=1}^{k} M_{m,R},$$

so that

$$\partial M_R = \bigsqcup_{m=1}^{k} S_{R,m}.$$

Applying Corollary 4.9 to ψ on the compact domain M_R, and using $\mathscr{D}\psi = 0$, we obtain

$$\int_{M_R} \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi \rangle \Big)\, \mu = \sum_{m=1}^{k} \int_{S_{R,m}} \beta(\psi). \tag{7.4}$$

By Proposition 3.39, the integrand on the left-hand side is pointwise nonnegative under the dominant energy condition. Therefore

$$\int_{M_R} \Big(|V\psi|^2 + \langle \psi, \mathcal{R}\psi \rangle \Big)\, \mu \geq 0 \qquad \text{for every } R. \tag{7.5}$$

We now pass to the limit as $R \to \infty$. Since ψ approaches the asymptotic constant spinor $\psi_{\infty,m}$ on each end, Proposition 4.23 gives

$$\lim_{R\to\infty} \int_{S_{R,m}} \beta(\psi) = 4\pi \Big(E_m \langle \psi_{\infty,m}, \psi_{\infty,m} \rangle + \big\langle \psi_{\infty,m}, (P_m)_k\, dx^0 \cdot dx^k \cdot \psi_{\infty,m} \big\rangle \Big).$$

Because of (7.1), all terms with $m \neq \ell$ vanish. Hence

$$\lim_{R\to\infty} \sum_{m=1}^{k} \int_{S_{R,m}} \beta(\psi) = 4\pi\Big(E_\ell \langle \psi_{\infty,\ell}, \psi_{\infty,\ell}\rangle + \big\langle \psi_{\infty,\ell}, (P_\ell)_k \, dx^0 \cdot dx^k \cdot \psi_{\infty,\ell}\big\rangle\Big). \tag{7.6}$$

Combining (7.4), (7.5), and (7.6), we obtain

$$0 \leq 4\pi\Big(E_\ell \langle \psi_{\infty,\ell}, \psi_{\infty,\ell}\rangle + \big\langle \psi_{\infty,\ell}, (P_\ell)_k \, dx^0 \cdot dx^k \cdot \psi_{\infty,\ell}\big\rangle\Big). \tag{7.7}$$

Thus the theorem has been reduced to a linear algebra statement on the asymptotic spinor space at the chosen end.

7.2. **Choice of asymptotic eigenspinor.** Recall from Proposition 4.25 that the asymptotic endomorphism

$$\mathcal{P}_\ell := (P_\ell)_k \, dx^0 \cdot dx^k$$

is Hermitian and satisfies

$$\mathcal{P}_\ell^2 = |P_\ell|^2 \, \mathrm{Id}.$$

Hence its eigenvalues are

$$\pm |P_\ell|.$$

Choose $\psi_{\infty,\ell} \neq 0$ to be an eigenvector of $\mathcal{P}_\ell$ with eigenvalue

$$\mathcal{P}_\ell \psi_{\infty,\ell} = -|P_\ell| \, \psi_{\infty,\ell}. \tag{7.8}$$

Then

$$\big\langle \psi_{\infty,\ell}, \mathcal{P}_\ell \psi_{\infty,\ell}\big\rangle = -|P_\ell| \, \langle \psi_{\infty,\ell}, \psi_{\infty,\ell}\rangle.$$

Substituting this into (7.7) gives

$$\begin{aligned} 0 &\leq 4\pi\Big(E_\ell \langle \psi_{\infty,\ell}, \psi_{\infty,\ell}\rangle - |P_\ell| \, \langle \psi_{\infty,\ell}, \psi_{\infty,\ell}\rangle\Big) \\ &= 4\pi(E_\ell - |P_\ell|) \, \langle \psi_{\infty,\ell}, \psi_{\infty,\ell}\rangle. \end{aligned} \tag{7.9}$$

Since $\psi_{\infty,\ell} \neq 0$ and the Hermitian form $\langle \cdot\,, \cdot \rangle$ is positive definite,

$$\langle \psi_{\infty,\ell}, \psi_{\infty,\ell}\rangle > 0.$$

Therefore (7.9) implies

$$E_\ell - |P_\ell| \geq 0. \tag{7.10}$$

7.3. **Conclusion.** The end M_ℓ was arbitrary. Hence (7.10) holds for every asymptotic end of M. We have therefore proved the positive energy theorem:

Theorem 7.11 (Positive energy theorem)**.** *Let $(N, g_{\alpha\beta})$ be a 4-dimensional Lorentz manifold satisfying Einstein's equation, and let $M \subset N$ be a complete oriented spacelike hypersurface whose induced initial data (M, g, h) are asymptotically flat. Assume the dominant energy condition holds along M. Then on each end M_ℓ,*

$$E_\ell \geq |P_\ell|.$$